A Deep Learning Approach for Wet Snow Monitoring in Mountainous Regions From SAR Image Time Series Based on Sentinel-1 and Sentinel-2 Snow Products

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***Abstract*—Snow is a vital environmental parameter that holds significance across various disciplines, such as hydrology, meteorology, and natural disaster management. With the increasing accessibility of snow products derived from synthetic aperture radar (SAR) and optical data, like Sentinel-1 wet snow (WS) and Sentinel-2 total snow, users have benefited from improved snow mapping and monitoring. However, snow mapping in the mountainous areas remains challenging due to the difficulty of obtaining reliable ground-truth data on steep mountain terrain. In this study, we introduce a deep semantic segmentation framework, separable atrous convolution U-Net (SACUNet), specifically designed for WS detection from SAR image time series in mountainous environments. To address the lack of ground truth, we constructed a high-confidence training and validation database through a rigorous decision-fusion process combining multitemporal Sentinel-1 WS detections with Sentinel-2 total snow maps. We also propose two complementary metrics, the conditional agreement rate (CAR) and the WS intersection over union (WSIoU), to quantify the robustness and consistency of the fusion procedure, thus ensuring the reliability of training labels in the absence of in situ data. SACUNet integrates advanced techniques like: 1) depthwise separable convolution (DSC), which captures cross-channel dependencies and adapts feature representations and 2) atrous separable convolution (ASC), which further refines and consolidates the learned features, into the U-Net architecture. The proposed framework has been successfully employed to monitor WS in the Mont-Blanc massif, using a time series of 69 Sentinel-1 images acquired from July 5, 2020 to August 29, 2021. SACUNet demonstrates remarkable accuracy in WS detection, with an overall accuracy (OA) of 97%, precision of 94%, recall of 97%, IoU at 92%, and an $F1$-score reaching 96%. Validation against meteorological records from four alpine stations confirmed that SACUNet effectively tracks seasonal WS dynamics, suppresses false detections during cold periods, and captures realistic high-altitude melt events. Moreover, the model trained in Mont-Blanc generalized successfully to the Vanoise massif, demonstrating its transferability to other alpine regions. Beyond quantitative accuracy, SACUNet enables the spatiotemporal analysis of WS evolution, offering insights into its extent, frequency, and seasonal progression across elevation bands. These findings highlight the framework's potential as an operational tool for large-scale WS monitoring in mountainous environments.**

***Index Terms*—Deep learning (DL), Sentinel-1 and Sentinel-2 snow products, separable atrous convolution U-Net (SACUNet), synthetic aperture radar (SAR) image time series, wet snow (WS) segmentation.**

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I. Introduction

SNOW profoundly impacts natural processes, human activities, and environmental stability. The significance of snow extends to various aspects of human life, including water resource management, climate research, and disaster mitigation. It serves as a vital natural reservoir, storing water as snowpack during winter, which gradually melts to provide essential freshwater downstream [1]. This water resource sustains ecosystems, agriculture, and hydropower generation. Moreover, snow in mountainous regions contributes to local climate regulation, influencing temperature patterns, indicating climate change impacts, and plays a pivotal role in avalanche risk assessment and disaster preparedness, safeguarding lives and infrastructure [2]. Detecting and monitoring snow, especially wet snow (WS), is crucial for managing water resources, ensuring public safety, comprehending climate dynamics, and preserving ecosystems. Snow and WS cover data are essential for research in climatology, hydrology, glaciology, and environmental science [3], [4], [5], [6].

Synthetic aperture radar (SAR) is very helpful to detect and monitor snow cover in various environments. Unlike optical data, SAR sensors can capture images through clouds and darkness, ensuring year-round monitoring even in adverse weather conditions. They provide high-resolution and frequent multivariate SAR data (i.e., multidate, multipolarization (PolSAR), interferometric SAR (InSAR), multifrequency, etc.) that

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is especially valuable for snow monitoring. Indeed, high spatial resolution SAR data enable the detection of small-scale snow cover changes [7], [8]. The availability of time-series SAR data allows advanced change detection and monitoring of snow cover dynamics, including snow accumulation, melting, and snowpack properties [9], [10], [11]. Besides, PolSAR data provide additional information on snow properties, enhancing the discrimination of WS, dry snow, and ice [12], [13]. In addition, InSAR techniques have been found helpful to measure snow depth changes over time by analyzing phase differences in SAR images, which is valuable for understanding snowpack thickness variations [14], [15], [16]. Moreover, multifrequency SAR data offer information about the physical properties of snow cover, including its density, wetness, and grain size. For instance, SAR at lower frequencies (e.g., L-band) can penetrate deeper into the snowpack due to longer wavelengths, allowing for the estimation of snow wetness and density. These estimates are critical for assessing snowmelt rates and the potential for flooding [17], [18], [19]. Microwave scattering at higher frequencies (e.g., C-band and X-band) is influenced by snow grain size and shape, which are essential for understanding snow metamorphism and metamorphic processes [20], [21], [22].

WS extent mapping and monitoring from SAR image time series involves analyzing SAR backscatter values to distinguish snow conditions. WS typically exhibits lower backscatter values compared to dry snow due to the presence of liquid water, which absorbs microwave energy [23], [24]. The study in [25] demonstrates that SAR data like Sentinel-1 can effectively monitor snowmelt dynamics by detecting changes in snow water content. By analyzing SAR backscatter across several alpine sites, the authors identified distinct snowmelt phases, i.e., moistening, ripening, and runoff, showing that SAR data are sensitive to meltwater release. A thresholding approach can be applied to the backscatter values to classify areas as dry snow, WS, or other land cover types [26]. Utilizing horizontal and vertical polarization SAR data can provide additional information for better distinguishing snow from other land cover types [27], [28], [29], [30], [31]. Change detection is another essential technique for WS detection from SAR images. It involves comparing two SAR images of the same scene acquired at different times to identify areas transitioning from snow-free, dry snow, or frozen ground conditions to WS [26], [32], [33], [34], [35], [36], [37], [38], [39]. To improve WS detection results, in [40], the local intensity ratio was considered as a stochastic process, and the variation of local incidence angle was taken into account. Besides, several similarity metrics were examined as change detectors (such as the normalized cross-correlation ratio, the Hausdorff distance, and some other more refined image distance functions) in [41]. The combination/fusion of SAR data with optical imagery provides complementary information and a comprehensive view of snow cover, thus enhancing WS mapping [42], [43].

Machine learning (ML)/deep learning (DL) algorithms have recently demonstrated their effectiveness across various facets of snow monitoring and analysis using SAR data. For snow cover mapping and classification, including distinguishing WS from dry snow, ML is frequently applied to extract information on scattering mechanisms within the snowpack derived from PolSAR-based approaches. In addition to the original PolSAR channels (i.e., HH, VV, HV, and VH), PolSAR decomposition products (e.g., Pauli, $H/A/\alpha$, Cloude–Pottier, Freeman–Durden, Yamaguchi, etc.) are often used to extract snow cover information [42], [44]. Besides, the combination of different SAR information such as backscattering coefficient, InSAR coherence, polarimetric parameters, and other auxiliary data like topographical and meteorological factors can help improve snow detection [45], [46]. Conventional ML algorithms such as support vector machines (SVMs) and random forest (RF) are widely adopted. SVM, a supervised classification method, is well-known for its ability to handle high-dimensional nonlinear classification tasks, even with limited samples [42], [46], [47]. On the other hand, RF is recognized for its simplicity and low computational overhead, rendering it an attractive option [45]. To address the lack of quantitative evaluation in detecting WS, the study in [48] proposes a supervised learning approach using Sentinel-1 SAR imagery and topographic data. By labeling data with the CROCUS physical snow model and testing various ML models, this study finds improved accuracy in detecting WS, especially during the melting season, and reduces misclassification in winter. ML models are also applied in various other aspects of snow monitoring using SAR data, including tasks such as snow depth retrieval and snow water equivalent estimation. For example, in [49], the extreme gradient boosting (XGBoost) algorithm was used to retrieve snow depth. The dataset included optical and passive microwave remote sensing data, along with Sentinel-1-derived snow depth. Furthermore, in [50], the XGBoost algorithm was also employed and compared with RF for the same task. In [51], snow depth and snow water equivalent were simultaneously estimated from X- and Ku-band snow SAR data using artificial neural networks (ANNs). These ANNs were trained using both experimental data (data-driven) and data simulated through dense medium radiative transfer. In [52], ANNs and support vector regression were applied to estimate snow water equivalent. With the improvement of computer performance, DL algorithms have become successful in image classification tasks. However, there have been few attempts utilizing the deep neural network to monitor snow from SAR data [29]. In [44], an unsupervised deep training network named neighborhood-based sparse autoencoder (NSAE)-weighted fuzzy C-means clustering (WFCM) was designed for discriminating between dry snow and WS based on polarimetric features extracted from the H–α decomposition of dual-Pol Sentinel-1 SAR data. The NSAE-WFCM employs pixel-level NSAE to optimize polarimetric parameters, and it inputs the reconstructed features into the feature-based WFCM, assigning varying weights to each of them. In [53], a DL model was adopted utilizing both DL features, extracted from spectral data, and hand-crafted features derived from SAR and elevation data for snow cover mapping. A fully convolutional neural network (CNN) that takes input from both Sentinel-1 SAR data and topographical features was employed in [54] for the segmentation of snow avalanches.

Snow mapping and monitoring in mountainous areas is complex due to the challenges in obtaining accurate ground-truth data. Conducting fieldwork in these areas can be risky and challenging, and collecting in situ measurements is labor-intensive and costly. Remote sensing solutions have

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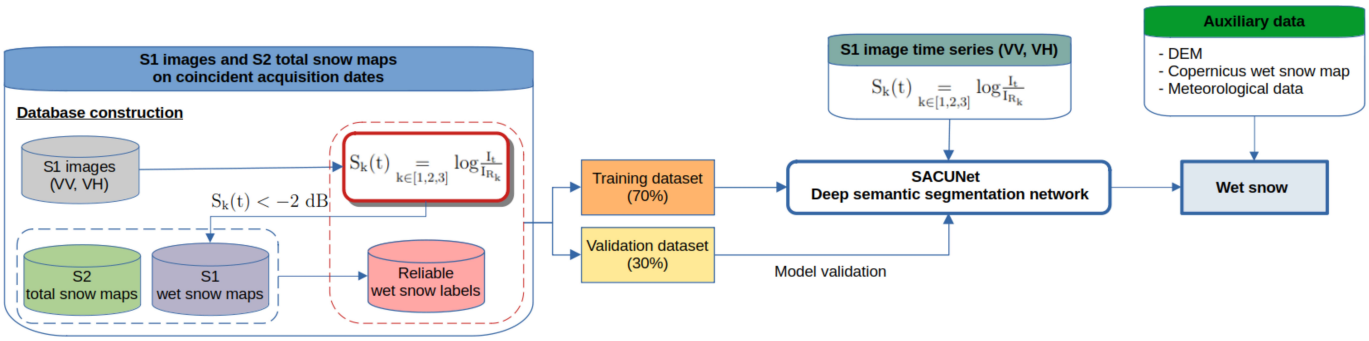


Fig. 1. General deep semantic learning framework for WS detection in SAR image time series.

become instrumental in mitigating these challenges, offering a wide-scale view of snow cover and enabling continuous and frequent monitoring. Numerous snow products derived from SAR and optical data are now readily available, such as Sentinel-2 total snow and Sentinel-1 WS, such as Sentinel-2 total snow developed by French National Centre for Space Studies (CNES) and the Center for the Study of the Biosphere from Space (CESBIO) from the THEIA Centre, and Sentinel-1 WS from Copernicus Land Monitoring Service. ML/DL methods can provide significant advantages over nonlearning approaches for snow/WS detection from remote sensing images. Unlike traditional methods that require manual feature engineering, ML/DL models can automatically learn complex patterns in remote sensing data, enhancing information representation. These approaches can better adapt to specific satellite image characteristics, such as speckle noise in SAR data, and can take advantage of large datasets for improved generalization. The automation of feature learning reduces dependence on human expertise, enhancing the efficiency of snow detection. Additionally, with the ability to extract intricate features from diverse data sources, ML/DL methods also hold potential for generating "total snow" products with a distinction between WS and dry snow through the integration of various satellite images, including SAR and optical data, and by fusing different available snow products, such as WS maps, total snow maps, and fractional snow products.

In this context, we introduce a deep semantic learning framework for the segmentation of Sentinel-1 images to detect WS in mountainous regions. The proposed framework can take advantage of SAR imagery and advanced DL techniques, offering a more accurate, efficient, and adaptable approach for WS extent mapping. Simultaneously, it can take advantage of accessible snow product sources to provide reference data for training and validation processes of the model, thus tackling the problem of constructing a database in high-relief regions, which was briefly described in [55]. The proposed CNN named separable atrous convolution U-Net (SACUNet) in this study extracts high-level features from SAR log-ratio images for WS segmentation. The SACUNet was trained using validated snow products derived from both Sentinel-1 and Sentinel-2 data, ensuring a robust foundation of reference data for both model training and evaluation purposes. Subsequently, the trained SACUNet model was deployed to detect WS within the Mont-Blanc massif in the Alps, utilizing a long SAR image time series comprising 69 ground range detected (GRD) VV + VH ascending Sentinel-1 images acquired from July 5, 2020, to August 29, 2021. The results achieved a notably high accuracy, supporting the sustained and reliable monitoring of WS evolution in alpine environments.

This article is organized as follows. Section II is dedicated to the proposed framework. Section III describes the study area and the data used for WS monitoring. Section IV presents the obtained results and discussion. Finally, conclusions and perspectives are given in Section V.

II. Deep Semantic Learning Framework for WS Segmentation in SAR Image Time Series

In response to the challenge and the complexity of gathering ground-truth data for snow mapping in mountainous regions, this study explores the integration of advanced DL algorithms with SAR data, taking advantage of existing Sentinel-1 and Sentinel-2 snow products. Fig. 1 depicts the outlined deep semantic segmentation framework designed for WS mapping. This framework comprises four primary components: the construction of a DL database, the development of a deep CNN, the subsequent phases of model training and validation, and ultimately, the application of the trained model for detecting WS in the SAR image time series.

A. Database Construction

We propose generating a DL dataset based on snow products derived from Sentinel-1 and Sentinel-2 data. This approach aims to obtain reliable WS labels by capitalizing on the combined observations from both types of data. It is important to highlight that throughout the observation period, we exclusively utilize Sentinel-1 and Sentinel-2 snow products derived from images acquired on the coincident acquisition dates for the two data types. During the observation period of this study from July 5, 2020, to August 29, 2021, there are 11 coincident acquisition dates, as shown in Table I. Indeed, for Sentinel-1 WS products used in this phase, we adopted the Nagler method [26] using three distinct reference images acquired on specific dates when WS was absent (see Table I). These reference images essentially establish a critical baseline for the detection of WS in the Sentinel-1 data. Hereafter, we use the term *GRD feature image* to refer to the aggregated VV and VH polarimetric channels of Sentinel-1 GRD magnitudes, derived using a multiplicative wavelet approach as described in [56]

$$\mathcal{I}_t(i,j) = (\mathrm{GRD}_t[\mathrm{VV}](i,j))^{\alpha} \times (\mathrm{GRD}_t[\mathrm{VH}](i,j))^{\beta} \quad (1)$$

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TABLE I

DESCRIPTION OF SENTINEL-1 DATA. (SENTINEL-1 AND SENTINEL-2 COINCIDENT ACQUISITION DATES ARE IN BOLD, AND REFERENCE (I.E., SNOW-FREE) DATES ARE UNDERLINED)

Specifications	Sentinel-1 data
Operator	European Space Agency (ESA).
Satellite	Sentinel-1A, B.
Launched date	April 03, 2014 & April 22, 2016.
Satellite orbit	Ascending, track 88.
Repeat cycle	6; 12 days.
Imaging frequency	C-band at 5.4 GHz.
Imaging mode	Interferometric Wide swath (IW).
Data product	Ground Range Detected (GRD) Level-1.
Resolution	10 m (processed by S1-Tiling).
Polarization	VH, VV.
Number of images	69 images.
Acquisition dates (YYYYMMDD)	20200705; 20200711; 20200717; **20200723**; 20200729; 20200804; 20200810; 20200816; **20200822**; 20200828; 20200903; 20200909; 20200915; 20200921; 20200927; 20201003; 20201009; 20201015; 20201021; 20201027; 20201102; 20201108; 20201114; **20201120**; 20201202; 20201208; 20201214; 20201220; 20201226; 20210101; 20210107; 20210113; **20210119**; 20210125; 20210131; 20210206; 20210212; **20210218**; 20210224; 20210302; 20210308; 20210314; **20210320**; 20210326; 20210401; 20210407; 20210413; **20210419**; 20210425; 20210501; 20210507; 20210513; **20210519**; 20210525; 20210531; 20210606; 20210612; **20210618**; 20210624; 20210706; 20210712; **20210718**; 20210724; 20210730; 20210805; 20210811; **20210817**; 20210823; 20210829.

where $\alpha = 0.25$ and $\beta = 0.75$ have been used to derive the results presented in this article.

We employed the log-ratio between Sentinel-1 GRD feature images (acquired on coincident dates) and reference GRD feature images with the aggregation of the two available VV and VH polarizations, thanks to a geometric wavelet transform [56], as highlighted by (1).

Subsequently, we obtained the reference Sentinel-1 WS maps by applying a threshold value of −2 dB to the calculated log ratios, which allow the detection of a decrease in the backscatter coefficient of WS surface as compared to situations when the surface is covered with dry snow or snow-free. In addition to the Sentinel-1 WS maps, the Sentinel-2 total snow maps provided by the THEIA Centre were used. The algorithm and scientific evaluation of these total snow products are presented in [57]. One can note that two general strategies are commonly considered when combining multimodal data in a DL context: *data fusion* (where the raw inputs are merged and fed into the model) and *decision fusion* (where independent detection outputs are merged to create labels). In this work, we relied solely on decision fusion to ensure high-confidence reference labels for training and evaluation. Indeed, for every coincident acquisition date of Sentinel-1 and Sentinel-2 data, we obtained *three* Sentinel-1 WS maps $\mathfrak{S}$ (based on three snow-free reference dates) along with *one* Sentinel-2 total snow binary map $\mathcal{O}$. A sample pixel is labeled as "WS" only if all three Sentinel-1 maps classify it as WS and the Sentinel-2 map indicates it as total snow. Otherwise, it is determined as "non-WS." This decision rule is intentionally stringent. First, it requires Sentinel-1 detections to be temporally consistent, thus limiting the influence of noise or artifacts in individual acquisitions. Second, it relies on Sentinel-2 only to confirm the presence of snow, whether wet or dry, through its spectral information. Sentinel-2 cannot introduce snow pixels on its own, and it can only validate or exclude detections already supported by Sentinel-1. As a result, the fused product emphasizes reliability by retaining only WS areas that are both temporally stable and spectrally consistent, while systematically discarding ambiguous or inconsistent cases. This design prevents false positives since pixels cannot be labeled as WS solely from Sentinel-2 information. Instead, the main source of error becomes false negatives, when true WS detected by Sentinel-1 is rejected due to missing or uncertain information in the Sentinel-2 mask (e.g., cloud cover, misclassified snow, or data gaps). Nevertheless, the number of labeled samples obtained through this rigorous fusion strategy remains large, ensuring that the constructed database is sufficiently rich and diverse to support the quality of both the training and validation phases of the DL model.

In the constructed database, the input feature $\mathcal{S} = (\mathcal{S}_{i,j,k}(t))_{i,j,k}$ denotes a spatiotemporal log-ratio subimage, where (i, j) and t refer, respectively, to the spatial and temporal variables; and k is an index associated with the reference dates used. Each subimage comprises three Sentinel-1 log ratios, defined by the following equation, between the considered image $\mathcal{I}_t$ and three reference images $\mathcal{I}_{R_k}, k \in \{1, 2, \ldots, K\}$, along with the corresponding label indicating the presence or absence of WS. A pixel of this subimage feature has the form

$$\mathcal{S}_{i,j,k}(t) = \log\left(\frac{\mathcal{I}_t(i,j)}{\mathcal{I}_{R_k}(i,j)}\right) \tag{2}$$

where $\mathcal{I}_t(i, j)$ is defined by (1). To construct a database with reliable labels, we applied a strict decision fusion rule: unanimity among the Sentinel-1 observers (across the three reference dates) combined with confirmation by the Sentinel-2 map at a coarser spatial scale. Specifically, a pixel of a subimage is labeled as follows:

1) WS if

$$\sum_{k=1}^{k_0} \mathfrak{S}\left(\mathcal{S}_{i,j,k}(t)\right) = k_0 \text{ and } \mathcal{O}_{i,j}(t) = 1 \tag{3}$$

2) non-WS if

$$\sum_{k=1}^{k_0} \mathfrak{S}\left(\mathcal{S}_{i,j,k}(t)\right) = 0 \text{ and } \mathcal{O}_{i,j}(t) = 0 \tag{4}$$

where $\mathfrak{S}$ denotes the Nagler's WS decision function yielding a binary output. Subimages, represented as three-channel tensors corresponding to the three reference dates, together with their pixel-level label maps, are then integrated in a deep segmentation learning database called $\mathfrak{D}$. For the training of the DL model, 70% of the chosen samples from the database are employed (dataset $\mathfrak{D}_T$), with the remaining 30% reserved for model validation (dataset $\mathfrak{D}_V$).

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B. SACUNet for Semantic Segmentation

In this study, we propose a CNN architecture named SACUNet, as illustrated in Fig. 2, to assign a corresponding and unique class label to each pixel of a subimage pertaining to $\mathfrak{D}_T$. The proposed network has the U-Net architecture [58] characterized by a U-shape, which consists of an encoder (contracting path) and a decoder (expanding path). This design enables the network to capture both local and global features from input images while preserving detailed spatial information. The SACUNet consists of the following components [see Fig. 2(a)]:

1) Input $\xrightarrow{B_1} J^0$: entry block;
2) $J^u \xrightarrow{E_e} J^{u+1}$ ($u \in \{0,1,2,3\}$; $e \in \{1,2,3,4\}$): depthwise separable convolution (DSC)-Encoder blocks;
3) $J^4 \xrightarrow{A} J^5$: bridge 1—atrous separable convolution (ASC) block;
4) $J^5 \xrightarrow{B_2} J^6$: bridge 2;
5) $J^{v+5} \xrightarrow{D_e} J^{v+6}$ ($v \in \{1,2,3,4\}$): decoder blocks;
6) $J^{10} \xrightarrow{B_3} J^{11}$: classification head;
7) $J^{11} \xrightarrow{\lambda}$ output: softmax.

From now on, the convention $\mathbf{H}_{N_f,s,d}^{M_1,M_2,C,N_g} = \bigcup_{m=1}^{N_g}\{\mathsf{H}_{N_f,s,d}^{M_1,M_2,C,m}\}$ is used to denote a convolution tensor composed by N_g convolution groups and N_f convolution filters $\mathsf{H}_{N_f,s,d}^{M_1,M_2,C} = \{\mathsf{h}_{n,s,d}^{M_1,M_2,C} : n = 1,2,\ldots,N_f\}$ per group, when any filter $\mathsf{h}_{n,s,d}^{M_1,M_2,C}$ has a spatial size $M_1 \times M_2$ and a channel size C, given that the convolution operation is associated with a stride s and a dilation factor d. We assume that the number of groups is 1 when group information is not provided.

1) Entry Block: The entry block prepares the input data for deeper layers in the network, which consists of the following:

1) a convolution layer associated with the convolution filter tensor $\mathsf{H}_{N,1,1}^{3,3,K}$;
2) a batch normalization operator;
3) a nonlinear activation function (chosen to be the rectified linear unit (ReLU) in this work).

The feature image resulting from the entry block is denoted as: $(J_{\bullet,\bullet,q}^0)_{1\leqslant q\leqslant N_0}$, see (11) for the detailed input–output relations yielding this feature image.

2) DSC-Encoder Blocks: In these blocks, more complex feature extraction and hierarchical processing occur. The block diagram on the left of Fig. 2(b) shows the encoder of the network. The encoder is responsible for progressively reducing the spatial dimensions of the input data while capturing context and extracting high-level features from the input data. For this contracting path, we employed the DSC technique, which is renowned for its efficiency and ability to reduce model complexity (e.g., Xception [59] and MobileNets [60] networks). DSC breaks down the standard convolution into two separate layers: depthwise convolution and pointwise convolution. Depthwise convolution applies a separate convolutional filter to each input channel (feature map) individually. It focuses on capturing spatial information within each channel while preserving channelwise independence. This step significantly reduces the number of parameters compared to standard convolution, while pointwise convolution, also known as 1×1 convolution, performs a traditional convolution operation but with a 1×1 kernel. It combines the spatial information from depthwise convolution across different channels. This step helps capture cross-channel dependencies and adapt feature representations.

The output J^{u+1} of a decoder block from the input J^u is given in the Appendix, see (12), showing that different DSCs are applied in each encoder block. Following the output of each DSC, there are nonlinear operators for batch normalization and ReLU activation. Subsequently, a MaxPooling operation is also applied to the feature image produced by the DSCs, effectively reducing spatial dimensions with a stride factor of 2. Simultaneously with the DSC process, each encoder block applies a pointwise convolution to the input, using a stride factor of 2. This parallel operation is strategically employed to bolster the network's capacity for feature extraction, encompassing both low-level and high-level features from the input data.

We utilized four encoder blocks to: 1) strike a balance between capturing the global context and preserving intricate details from input patches, which were relatively small in size (e.g., 80×80 pixels) and 2) focus on small variations and minimize the degradation of output resolution (i.e., avoid smoothing too much the segmented snow map).

The output size of each encoder block is halved in each spatial dimension compared to the input image, and the pointwise convolutions have doubled the numbers of filters when passing from block to block. Here, ends the encoding part, where we began with an input subimage having $K = 3$ channels and ended by describing the spatio-channel information contained in this subimage into a feature map having N_{U+1} channels.

3) Bridge 1—ASC Block: In order to further refine and consolidate the learned features before passing them to the decoder for upsampling and generating the final segmentation map, we employed the ASC. Atrous convolution introduces a dilation rate parameter, denoted as d, which controls the spacing between the values in a convolutional kernel. Unlike standard convolution, where d equals 1, atrous convolution involves spacing the kernel values by a certain rate, typically greater than 1. This dilation rate effectively expands the receptive field of the convolutional layer. It allows the network to capture features at different scales and hierarchies, making it particularly valuable for handling objects or structures of varying sizes within an image. A standard atrous convolution layer filters a sample $\mathcal{S}$ from different dilatation factors and concatenates the obtained output features.

The ASC is a composite convolution technique that combines depthwise, pointwise, and atrous convolutions. This approach was first introduced in Deeplabv2 [61] and subsequently refined in DeeplabV3+ [62]. It addresses a critical challenge in image segmentation by enabling the capture of multiscale information, thereby enhancing segmentation accuracy. In [62], one 1×1 convolution layer was coupled with three 3×3 convolution layers, each employing different dilation rates $d \in \{6,12,18\}$ for processing larger input sizes. For scenarios involving smaller input sizes, such as the ones considered in this work, we made adjustments to optimize feature extraction. Specifically, we utilized one 1×1 convolution layer along with three 2×2 convolution layers, with dilation rates $d \in \{2,3,4\}$. This is well-suited for efficiently extracting features from smaller input dimensions.

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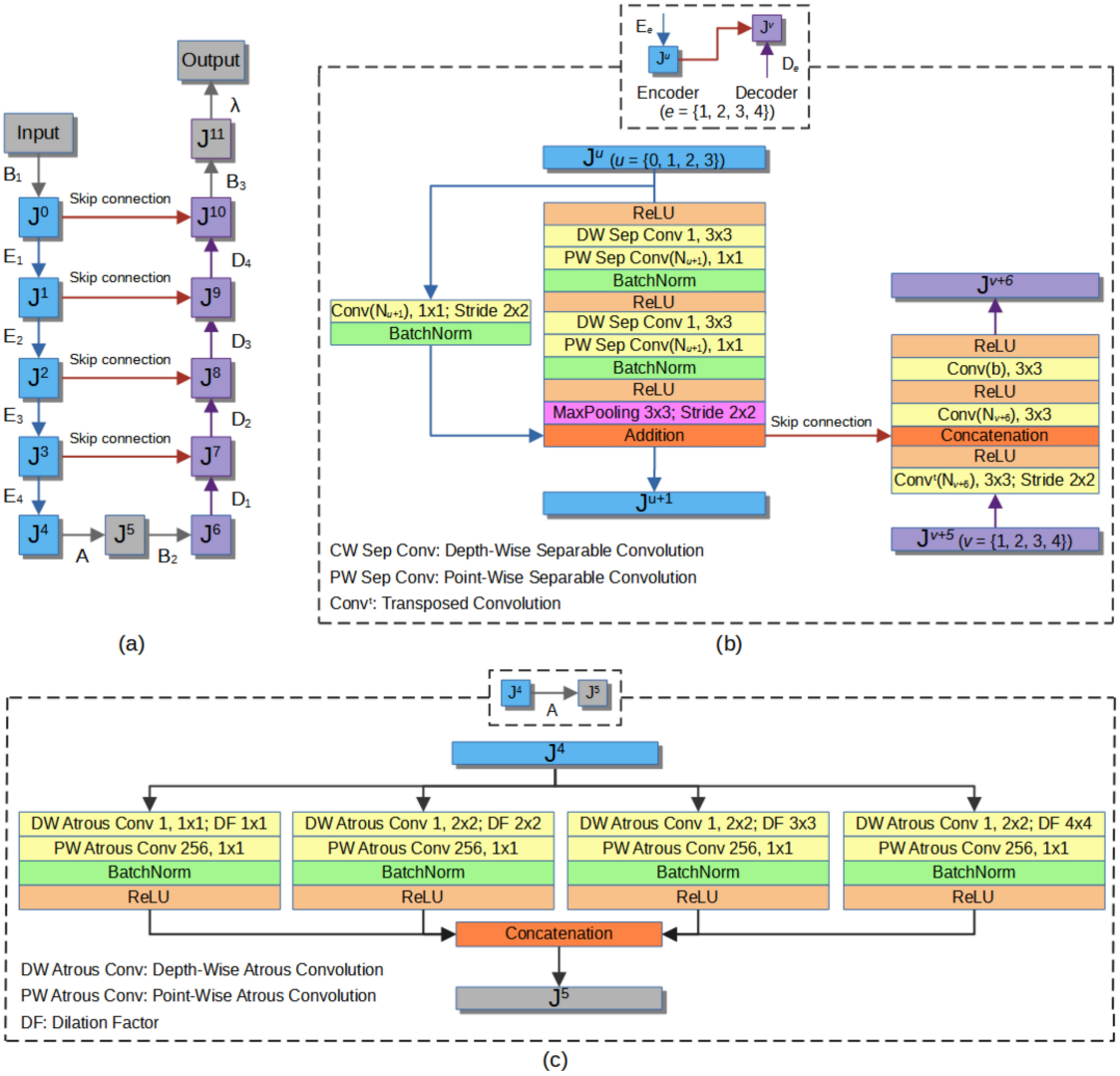


Fig. 2. SACUNet architecture. (a) Overview of the proposed SACUNet. (b) DSC-Encoder, decoder, and skip connections. (c) Multiscale ASC.

Fig. 2(c) shows the atrous block, which allows the feature extraction from multiple scales of information. These extracted features are then combined through concatenation to form a comprehensive set of learned features, encompassing a wide range of rich and diverse information. The feature image of the atrous block is denoted as $(J^{U+2}_{\bullet,\bullet,q})_{1\leqslant q\leqslant N_{U+2}}$ and its equation is given in the Appendix [see (13)].

4) Bridge 2: A convolution tensor with a spatial filter size of 3×3 and an ReLU function are then used in this block to further refine learned features before passing them to the expansive path for upsampling and generating the final segmentation map. The output of this step is: $(J^{V}_{\bullet,\bullet,q})_{1\leqslant q\leqslant N_{U+3}}$, whose analytical expression is given in (14). From now on, let $V = U + 3$.

5) Decoder Blocks: The block diagram on the right of Fig. 2(b) shows layers used in each decoder block. These blocks aim to reverse the downsampling and spatial dimension reduction performed by the encoder, gradually increasing the feature map's spatial resolution to match the input image size. We utilized transposed convolutions, which are particularly effective as they can learn the upsampling process. The output of each decoder block for $v \in \{1, 2, \ldots, U\}$ is denoted as $(J^{V+v}_{\bullet,\bullet,q})_{1\leqslant q\leqslant N_{V+v}}$ and is expressed in (15), where $V = U + 3$.

Specifically, in the derivation of this feature image, the decoder reconstructs spatial detail and refines the segmentation by applying the following operations:

1) Skip connections are used in this path to improve the accuracy of segmentation and maintain important details. These connections connect the decoder's layers with the corresponding layers from the encoder.
2) We used a concatenation layer to combine features along the channel dimension between the upsampled image generated by the transposed convolution layer and the corresponding output of the addition layer of the encoder.
3) The result of a skip connection undergoes further refinement through two successive convolutional layers, each followed by an ReLU activation, in preparation for the subsequent stages, where the output size of each encoder block is doubled compared to that of the input image.
4) By merging features from both the shallow and deep layers, the decoder can combine high-level semantics with low-level details, resulting in more accurate and visually appealing outputs.

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It is important to highlight that to mitigate challenges associated with employing a deep neural network originally tailored for larger input images (e.g., 80 × 80 pixels) on a dataset comprising smaller input images (e.g., 40 × 40 pixels or 20 × 20 pixels), including potential issues such as the loss of spatial information, increased computational demands, overfitting risks, and reduced generalization capabilities, the number of encoder and decoder blocks (typically set to four blocks by default) can be modified to align with the dimensions of the input data. This adjustment allows for more effective feature extraction and model performance when dealing with smaller sized images.

6) Classification Head: In this block, the convolutional layer takes on the role of further processing and extracting relevant features from the output of the decoder block. Its purpose is to prepare the image, which maintains the same spatial size and comprises two channels representing the two classes of WS and non-WS, in preparation for the subsequent softmax layer. The output feature is denoted as $(J_{\bullet,\bullet,q}^{V+U+2})_{1\leqslant q\leqslant N_{V+U+2}}$, where $N_{V+U+2} = 2$ according to the desired binary output. The input–output relation is given in the Appendix [see (16)].

7) Softmax: The network estimates for each class $q \in \{1, 2\}$ its softmax categorical probability as follows:

$$\widehat{p_{i,j,q}} = \frac{e^{J_{i,j,q}}}{e^{J_{i,j,1}} + e^{J_{i,j,2}}} \tag{5}$$

and updates its convolutional weights, thanks to the behavior of the cross-entropy loss. Taking into account the different subimages in $\mathfrak{D}_T$, then the loss function is

$$\mathfrak{H} = -\frac{1}{\text{Const}} \sum_{k:I_k\in\mathfrak{D}_T} \sum_{i,j} \sum_{q=1,2} p_{i,j,q}(I_k) \log \widehat{p_{i,j,q}(I_k)} \tag{6}$$

with $\{p_{i,j,1}, p_{i,j,2}\}$ being the binary labels: $p_{i,j,1} + p_{i,j,2} = 1$ and "Const" is the product between the number of subimages in $\mathfrak{D}_T$ and the size of a subimage. The final decision, after the training step, is, thus, a pixelwise set of estimated labels with respect to the comparison

$$\widehat{p_{i,j,1}} \underset{\text{Non-WS}}{\overset{\text{WS}}{\gtrless}} \widehat{p_{i,j,2}}. \tag{7}$$

C. Model Training and Validation

We used an initial number of convolution filters $N = 64$ and $U = 3$, and thus, $V = 6$ in the experimental results: this corresponds to the block diagram of Fig. 2(a), noting that the same architecture is consistent for all $U \geqslant 1$.

1) Model Training: The SACUNet model, proposed in this study, undergoes comprehensive training using three distinct databases, each comprising subimages of varying sizes: 20 × 20, 40 × 40, and 80 × 80 pixels. The careful construction of these databases ensures the availability of an ample number of samples essential for the effective training and validation of the network. Specifically, these databases contain 83 157, 82 226, and 80 372 samples, respectively, for the three different sizes mentioned above. To validate the model, 30% of these samples were set aside for this specific purpose.

In the pursuit of optimal performance, several experiments were conducted to fine-tune the hyperparameters governing the model's behavior. The chosen hyperparameter settings for this study are as follows: an initial learning rate of 0.001, a momentum of 0.9, a batch size of 32, a maximum training epoch limit of 20, and the utilization of the stochastic gradient descent (SGD) optimization algorithm. These hyperparameter selections emerged from a thorough process of experimentation and careful consideration, aiming to attain the highest achievable level of performance and accuracy in the model's WS mapping task. All experimental works were implemented on a computing server equipped with an NVIDIA RTX A6000 GPU.

2) Model Validation: For the quantitative evaluation of the decision fusion process between Sentinel-1 and Sentinel-2 snow products, we introduce two complementary statistical metrics: the conditional agreement rate (CAR) and the WS intersection over union (WSIoU). These metrics aim to explicitly identify potential failure modes in the label generation process and to assess the consistency and robustness of the training labels generated through fusion.

CAR quantifies the proportion of WS pixels consistently detected by all Sentinel-1 acquisitions that are also confirmed by the Sentinel-2 total snow extent, before fusion. This metric assesses how much the Sentinel-1 consensus aligns with the independent optical evidence from Sentinel-2, which serves as a soft constraint in our fusion logic. Although not a ground-truth validation, CAR provides insight into the internal consistency of the fusion process and allows us to identify potential failure modes (e.g., overestimations by Sentinel-1 or exclusion of valid Sentinel-1 WS due to Sentinel-2 limitations such as cloud cover or resolution). CAR is defined as

$$\text{CAR} = \frac{\sum_{i,j}\left(\prod_{k=1}^{K-1} D_k(i,j) \cdot D_K(i,j)\right)}{\sum_{i,j}\left(\prod_{k=1}^{K-1} D_k(i,j)\right)} \tag{8}$$

where (i, j) denotes the pixel location, D_k for $k \in [1, K-1]$ corresponds to Sentinel-1 WS maps, and D_K is the Sentinel-2 total snow map.

WSIoU measures the degree of agreement among the snow detection maps involved in the fusion process. Specifically, it evaluates the IoU between all contributing snow maps (three Sentinel-1 WS maps and one Sentinel-2 total snow map), ensuring that only pixels with strong consensus are retained as reliable WS labels

$$\text{WSIoU} = \frac{\sum_{i,j}\prod_{k=1}^{K} D_k(i,j)}{\sum_{i,j}\left(\sum_{k=1}^{K} D_k(i,j) > 0\right)} \tag{9}$$

where D_k represents the binary classification map indicating the presence of WS or total snow in the kth snow map among the total K maps used ($K = 4$ in this case). Together, CAR and WSIoU offer a robust, quantitative framework to validate the fusion strategy by measuring both the agreement across sources and the extent of cross-sensor consistency. Their joint analysis addresses concerns regarding the validity of the fusion process and helps interpret its limitations in the absence of in situ ground-truth data.

For the evaluation of the model's performance, we employ several key metrics, namely, precision, recall, $F1$-score, IoU, and overall accuracy (OA). These metrics are computed as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

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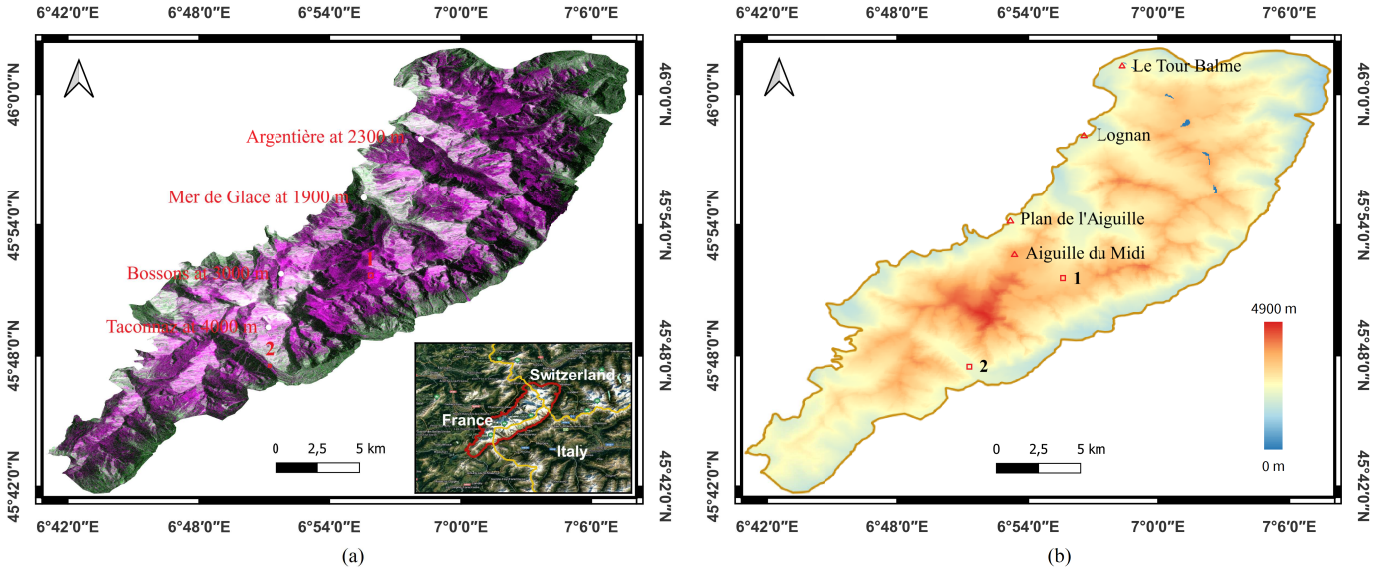


Fig. 3. Location of the study area, Mont-Blanc massif. (a) General map and RGB composition SAR image showing WS cover (in magenta) on June 6, 2021: R—January 19, 2021, G—June 6, 2021, and *B*—January 19, 2021; white circles: glacier observation areas in Fig. 16. (b) Elevation of the Mont-Blanc massif; red triangles: locations of meteorological stations: Lognan—1970 m, Plan de l'Aiguille—2250 m, and Aiguille du Midi—3845 m; (1) and (2): observed areas for calculating correlation coefficients among images within the time series, as shown in Fig. 11.

$$\begin{aligned} \text{Recall} &= \frac{\text{TP}}{\text{TP}+\text{FN}} \\ F1\text{-score} &= \frac{2\times\text{Precision}\times\text{Recall}}{\text{Precision}+\text{Recall}} \\ \text{IoU} &= \frac{\text{TP}}{\text{TP}+\text{FP}+\text{FN}} \\ \text{OA} &= \frac{\text{TP}+\text{TN}}{\text{TP}+\text{FP}+\text{FN}+\text{TN}} \end{aligned} \tag{10}$$

where TP, FP, TN, and FN are the true positive, false positive, true negative, and false negative, respectively.

The performance of the proposed SACUNet was also compared to five other DL models including the fully convolutional network (FCNet) [63] preinitialized using layers and weights from the VGG-16 network [64], the SegNet [65], the U-Net [58], and the DeeplabV3+ approach [62], which was implemented with two different backbones, ResNet18 [66] and Xception [59]. Furthermore, we conducted a comparative analysis of the WS extent over the Mont-Blanc massif obtained through our proposed framework with WS maps from the Copernicus Land Monitoring Service.

III. Study Area and Data Used

We conducted experimental work, in this article, in two selected massifs in the Alps. The primary focus was on the Mont-Blanc massif, prominently situated in the Western Alps, spanning the borders of France, Italy, and Switzerland (see Fig. 3). This massif features 11 main summits exceeding 4000 m and is home to iconic glaciers such as the Mer de Glace, Glacier des Bossons, and Glacier du Miage. Unfortunately, climate change has led to the rapid retreat of these glaciers, posing severe implications for the hydrology and ecosystems of the area. The Mont-Blanc massif experiences a blend of Mediterranean and Alpine climates, resulting in unique weather patterns [67]. Temperatures can fluctuate significantly, with cold winters and mild summers. Heavy snowfall during winter can persist for extended periods, contributing to the region's renowned snow-covered landscapes. However, the steep terrain and heavy snowfall pose significant avalanche risks, necessitating diligent monitoring and safety measures.

The Vanoise massif was selected as a test site to verify the applicability of the SACUNet model trained by databases of the Mont-Blanc study area to other regions. Located in the Graian Alps of southeastern France, the Vanoise massif is characterized by diverse natural conditions. The massif is known for its rugged and mountainous terrain, including numerous peaks, deep valleys, and high plateaus, with some prominent peaks in the region exceeding 3000 m. This region is also home to several glaciers although their extent has been reduced due to climate change. A significant portion of the massif is the Vanoise National Park, known for its rich biodiversity.

In this study, the WS extent of the Mont-Blanc massif from July 5, 2020, to August 29, 2021, was mapped using a Sentinel-1 ascending, track 88 SAR image time series. Table I summarizes the principal parameters and acquisition dates of the dataset used. The Sentinel-1 mission comprises two satellites, namely, Sentinel-1A (launched on April 3, 2014) and Sentinel-1B (launched on April 22, 2016), strategically positioned 180° apart, both equipped with the latest generation radar instruments. Unfortunately, Sentinel-1B concluded its mission on August 3, 2022. Sentinel-1C was subsequently launched on December 5, 2024, restoring the two-satellite configuration. Operating at a C-band frequency of 5.405 GHz, the Sentinel-1 SAR sensor supports both single- and dual-polarization modes, providing Earth observation data every six days (combining Sentinel-1A and B).

In addition to the Sentinel-1 data, we collected as many 20-m resolution Sentinel-2 total snow maps as possible from the THEIA Centre for the DL database creation purpose. These total snow maps were derived from Sentinel-2 images acquired on the same dates as the Sentinel-1 images in the time series.

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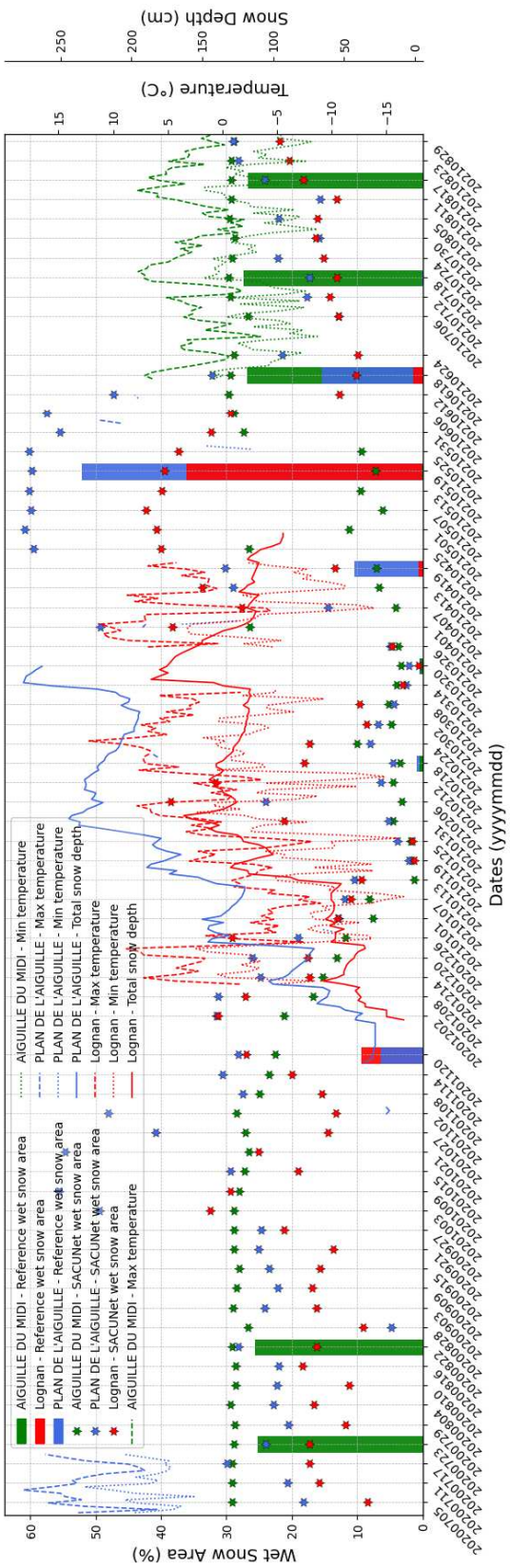


Fig. 4. Meteorological data and WS area detected by reference snow maps and SACUNet predictions across three observation stations (Aiguille du Midi, Plan de l'Aiguille, and Lognan) over time in the Mont-Blanc massif during the observation period from July 5, 2020, to August 29, 2021.

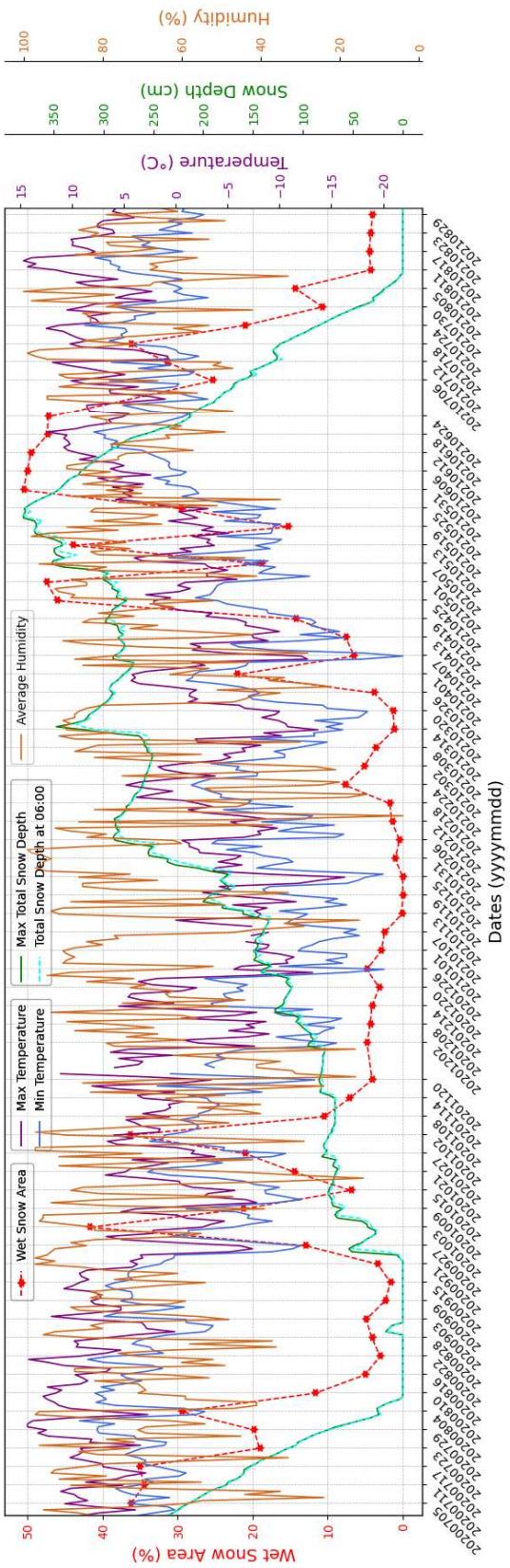


Fig. 5. Meteorological data and WS area detected SACUNet predictions at Bellecôte observation station over time in the Vanoise massif during the observation period from July 5, 2020, to August 29, 2021.

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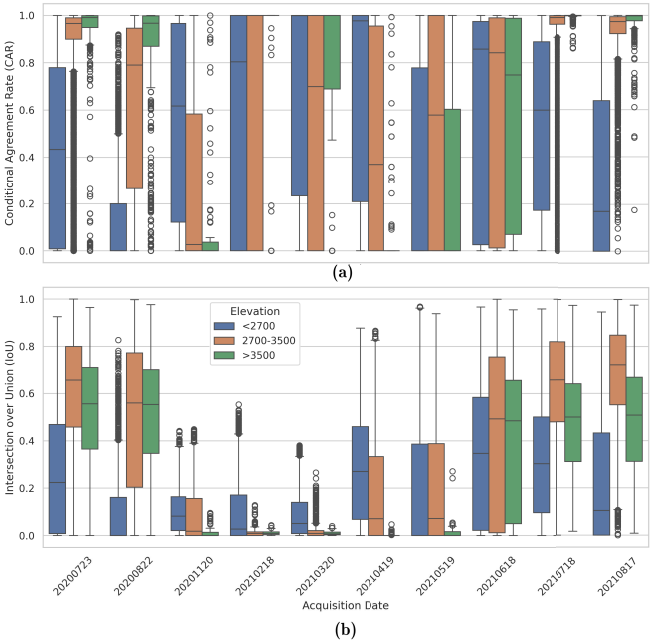


Fig. 6. Boxplots of (a) CAR and (b) WSIoU per coincident Sentinel-1/Sentinel-2 acquisition date, stratified by elevation classes (<2700, 2700–3500, and >3500 m). Higher values of CAR and WSIoU indicate stronger consistency and agreement across sensors (1—perfect agreement and 0—no overlap), whereas lower values reveal discrepancies or potential fusion failures.

The 11 coincident acquisition dates of Sentinel-1 and sentinel-2 data are shown in Table I. Moreover, meteorological data were also collected in both the study and test areas to support the analysis. The dataset used in the Mont-Blanc massif (see Fig. 4) includes temperature (maximum and minimum) and snow depth measurements recorded at three stations situated at different elevations. Aiguille du Midi (3842 m) provides only temperature data from June to August 2021, while Plan de l'Aiguille (2317 m) and Lognan (1970 m) provide both temperature and snow depth measurements, with data predominantly available between December 2020 and April 2021. In the Vanoise test site (see Fig. 5), meteorological data were obtained from the Bellecôte station, which offers year-round temperature (maximum and minimum), snow depth measurements (both maximum and values recorded at 06:00), and average humidity levels. These datasets provide essential reference conditions for validating WS detection. Furthermore, we also used 60-m resolution Copernicus Sentinel-1 WS maps corresponding to every SAR image in the time series, and a 10-m resolution digital elevation model (DEM) produced by French Institute of Geographic and Forest Information (IGN) for comparison and interpretation purposes.

Before WS mapping and analysis, the data preprocessing was implemented. For the SAR image time series, we adopted the S1Tiling tool. S1Tiling is a Python processing chain developed by CNES radar service, in collaboration with the CESBIO, based on the Orfeo ToolBox (OTB) image processing library [68]. The S1Tiling tool can automatically perform the following tasks: data downloading [from the Sentinel product exploitation platform (PEPS)], radiometric terrain correction, calibration, orthorectification, and speckle filtering. Preprocessed images are orthorectified on the Sentinel-2 grid and have the pixel size of 10×10 m [69]. To ensure consistent comparisons, Copernicus Sentinel-1 WS and Sentinel-2 total snow maps were resampled to match the dimensions of the Sentinel-1 data.

IV. Results

A. Model's Performance Evaluation

1) Evaluation of Fusion-Based Label Generation: Fig. 6 presents boxplots of CAR and WSIoU, computed patchwise for three elevation classes (<2700, 2700–3500, and >3500 m) and each coincident Sentinel-1/Sentinel-2 acquisition. Together, these metrics describe the cross-sensor consistency and internal agreement of the fusion process. During the summer acquisitions (July 23, 2020, and August 22, 2020; July 18, 2021, and August 17, 2021), CAR values were generally high (>0.8) at mid and high elevations, confirming strong coherence between Sentinel-1 WS and Sentinel-2 total snow, but dropped below 0.5 at <2700 m, where snow was mostly absent. This pattern reflects the physical conditions in Mont-Blanc-minimal snow at low elevations but persistent high-altitude melting detected by Sentinel-1. WSIoU values during these months were moderate (0.4–0.6), indicating good overall agreement, with minor temporal or resolution-induced mismatches among maps. In autumn (November 20, 2020), both metrics decreased sharply (CAR < 0.3 and WSIoU < 0.2) as snowpack stabilized and WS detections became sparse and heterogeneous, reducing overlap between radar and optical sources. In winter (February 18, 2021), CAR medians were close to 1.0 at 2700–3500 and >3500 m, and still >0.6 below 2700 m. Wherever Sentinel-1 consistently detected WS across its three maps, those pixels almost always coincided with snow-covered areas in Sentinel-2, showing strong cross-sensor coherence. WSIoU, however, remained near zero (occasional outliers up to 0.2), reflecting the rarity of WS under widespread dry conditions. Consistent Sentinel-1 detections were localized and seldom confirmed by all maps. In early spring (March 20, 2021), CAR remained high across elevations, but WSIoU was still near zero. By April 19, 2021, CAR stayed high, while WSIoU rose slightly, reflecting spatially fragmented and intermittent melt. From May 19 onward, CAR reached near saturation at mid and high elevations, and WSIoU increased steadily, marking a growing consensus among maps during active melting. By June 18, 2021, both metrics peaked, CAR near 1.0 and WSIoU higher and more compact, indicating true peak-melt agreement across all sources. Overall, these results confirm that the fusion process effectively removes inconsistent detections and retains only pixels jointly validated by temporally stable Sentinel-1 and Sentinel-2 data, ensuring spatial reliability suitable for model training and validation without ground truth.

Fig. 7 illustrates the stepwise fusion procedure used to construct reliable WS labels on July 23, 2020. Starting with single Sentinel-1 WS detections [Fig. 7(a)], numerous noisy or isolated patches are observed. Pairwise fusion between Sentinel-1 maps [Fig. 7(b)] removes many of these spurious detections, retaining only areas confirmed across acquisitions. Incorporating the third Sentinel-1 map [Fig. 7(c)] further enforces temporal consistency, ensuring that only pixels consistently classified as WS across all three radar acquisitions are preserved. This unanimity rule not only reduces false

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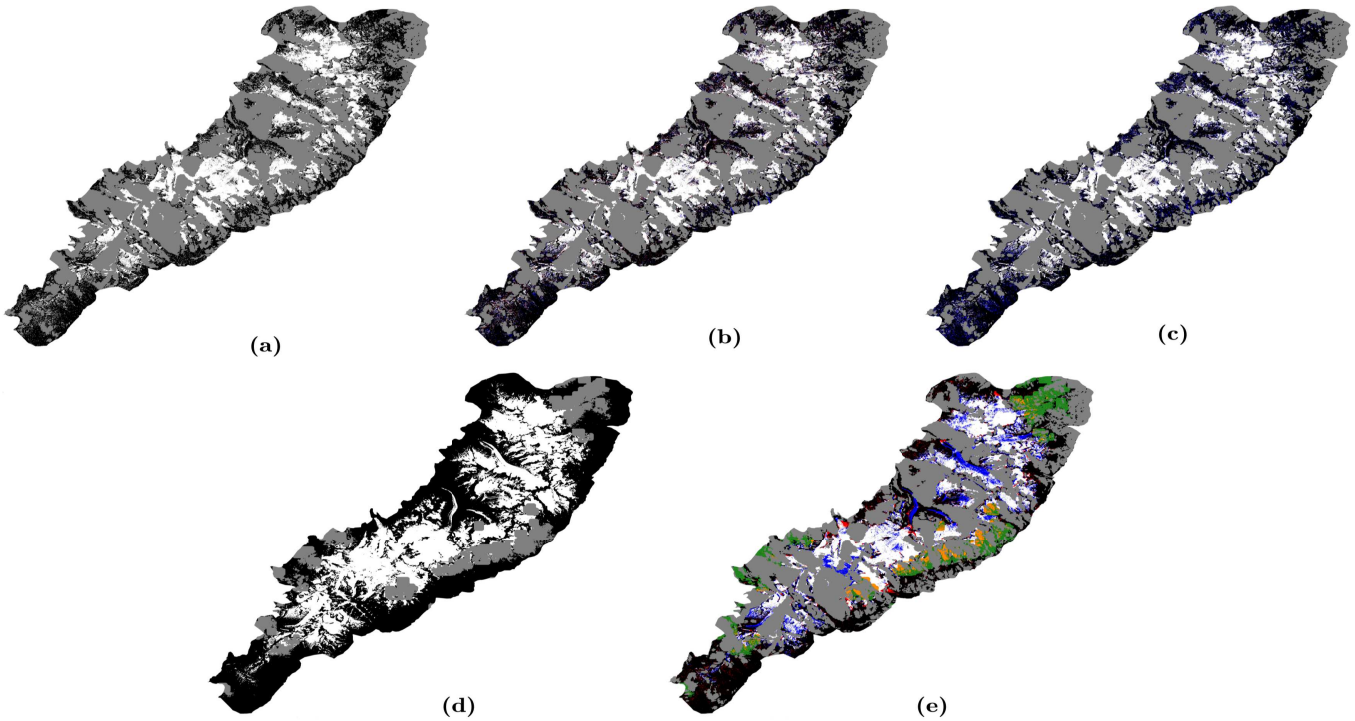


Fig. 7. Example of the label generation process on July 23, 2020. (a)–(c) Temporal fusion of three Sentinel-1 WS maps progressively removes inconsistent detections. (d) Sentinel-2 snow map further constrains the fusion. (e) Final fusion map showing reliable detections and explicit rejection classes. The same color scheme is applied across all steps: white—reliable WS retained; black—reliable non-WS retained; red/blue—WS (red) or non-WS (blue) excluded due to temporal or cross-sensor inconsistency; yellow/green—WS (yellow) or non-WS (green) excluded due to cloud/gap in the Sentinel-2 total snow map; and gray—undefined or invalid areas. Note: For all Sentinel-1 WS maps, nonmountainous areas were masked using Copernicus DEM GLO30 and CORINE Land Cover 2018, and radar-specific geometric distortions (foreshortening, foldover, and shadow) were also masked.

positives, but also rejects intermittent or short-lived WS events. The Sentinel-2 total snow map [Fig. 7(d)] provides an independent constraint that further refines the classification. By confirming Sentinel-1 WS detections against Sentinel-2 total snow coverage, the fusion process eliminates physically inconsistent cases (e.g., WS outside total snow-covered areas). However, cloud cover and data gaps in Sentinel-2 can lead to valid Sentinel-1 detections being discarded. The final fusion map [Fig. 7(e)], thus, combines the robustness of temporally consistent radar detections with the independent confirmation of optical snow extent. This stepwise analysis highlights both the successes and limitations of the decision fusion strategy. Successes include the suppression of isolated false positives and the retention of stable WS signals. Limitations appear in three main contexts. First, at snowline boundaries, the transition zones between snow-covered and snow-free terrain, pixels often contain a mixture of snow and bare ground. In these mixed-pixel conditions, Sentinel-1 may register radar backscatter consistent with WS, while Sentinel-2 may classify the same pixel as partially or completely snow-free, creating disagreements in the fusion process. Second, in rugged terrain, differences in acquisition geometry amplify misalignments between sensors. In snowline regions, even small spatial offsets or differences in pixel footprints can lead to conflicting classifications, causing valid WS detections to be discarded during fusion. Third, cloud cover and data gaps in Sentinel-2 introduce additional sources of rejection. In such cases, WS detected consistently by Sentinel-1 may be excluded when Sentinel-2 observations are missing or unreliable, which can lead to an underrepresentation of WS in cloudy regions. Overall, despite these limitations, the final fusion emphasizes spatial reliability by retaining only pixels jointly validated across radar and optical sources, thereby producing a training dataset well-suited for DL applications in the absence of in situ ground truth.

2) Meteorological Validation of the Generated Labels: To assess the reliability of WS labeling for SACUNet training and evaluate its classification accuracy, the WS training samples (over 11 coincident Sentinel-1, Sentinel-2 acquisition dates) and WS areas detected by SACUNet over 69 observation dates, within a patch of size 80×80 pixels, were compared against meteorological observations from three stations at different elevations in the Mont-Blanc massif: Aiguille du Midi (3842 m), Plan de l'Aiguille (2317 m), and Lognan (1970 m). Correlating the collected meteorological data with the detected WS results presents challenges due to data discontinuities throughout the observation period and limited parameter diversity. The dataset primarily includes temperature and snow depth measurements, lacking additional meteorological variables such as humidity, radiation, and wind speed, which can significantly influence WS formation and detection. These limitations restrict the ability to fully assess the relationship between atmospheric conditions and WS dynamics. The reference WS labels, represented as bars in Fig. 4, were assessed based on their consistency with temperature conditions near 0°C and the presence of a sufficient snowpack. It can be observed that WS was correctly labeled during periods of expected melting. At higher elevations (Aiguille du Midi, 3842 m), WS was

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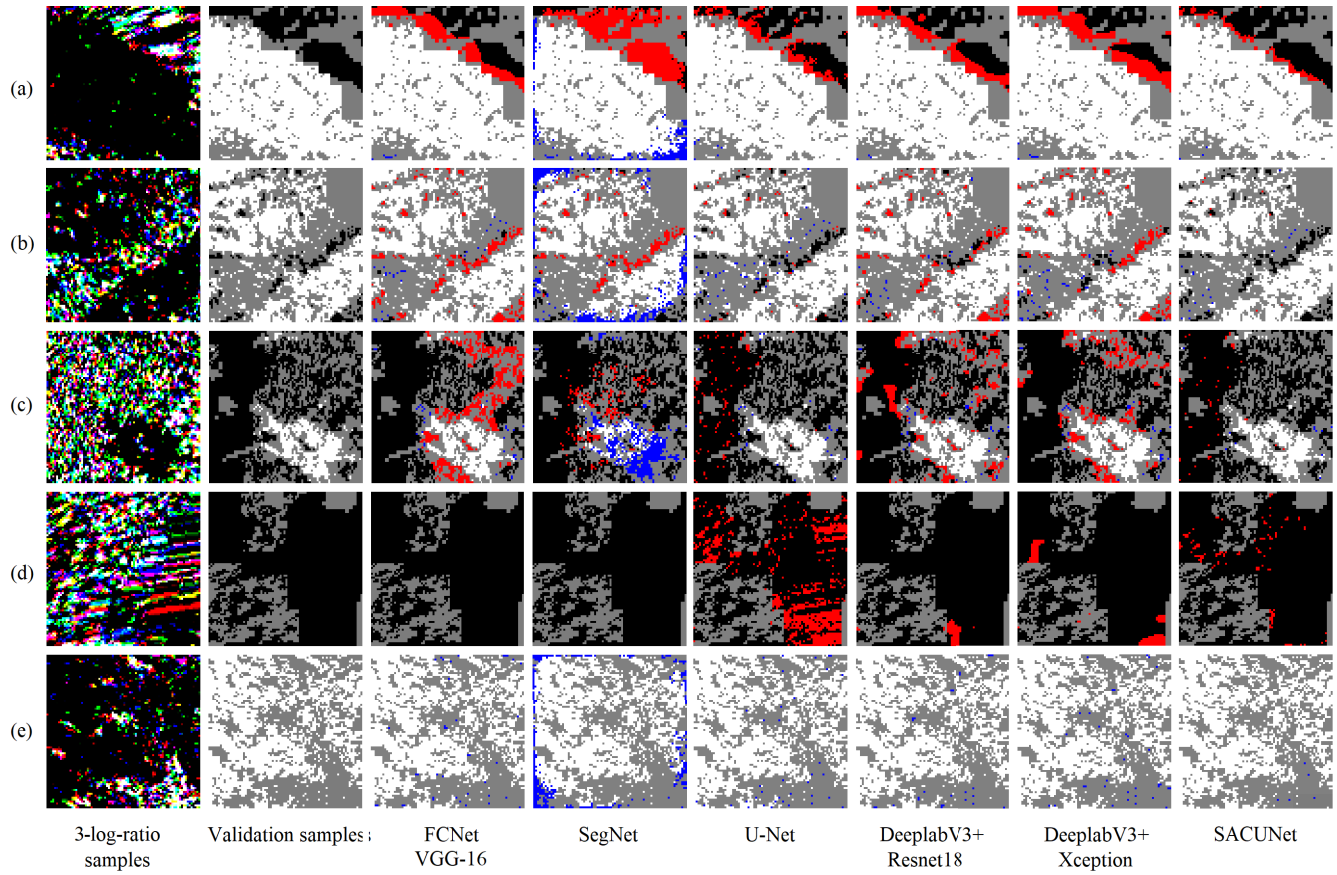


Fig. 8. WS detection performance of different models. (a)–(e) Five different test samples selected to illustrate model behavior across a variety of scene configurations, ranging from extensive wet-snow areas to more fragmented or heterogeneous patterns. 3-log-ratio samples: RGB composition [R: $\log(\mathcal{I}_t/\mathcal{I}_{\text{ref}_1})$, G: $\log(\mathcal{I}_t/\mathcal{I}_{\text{ref}_2})$, and B: $\log(\mathcal{I}_t/\mathcal{I}_{\text{ref}_3})$]; validation samples: white—reliable WS, black—reliable non-WS, and gray—undefined pixels; WS detection samples: white—true positive, black—true negative, red—false positive, and blue—false negative.

labeled only in summer (July–August 2020 and 2021), where temperatures were highest. The rarity or absence of WS labels in winter and spring at all three stations is consistent with its persistently subzero temperatures, confirming the reliability of the labeling. At mid-altitude station, Plan de l'Aiguille, 2317 m), WS labels appeared in May–June, August 2020, and August 2021, matching periods when temperatures rose above freezing, allowing for surface melting. Similarly, Lognan (1970 m), the lowest station, exhibited the highest number of labeled WS days in May–June. Notably, on January 19, 2021, one of the three selected reference dates, no WS was labeled due to the careful selection process applied in the database construction methodology, ensuring that only reliable WS observations were included. Across all stations, WS labeling correctly corresponds to periods of melting conditions and does not appear when subzero temperatures persist. This confirms that the Sentinel-1 and Sentinel-2 snow product-derived WS labels used for SACUNet training accurately represent actual WS occurrences, minimizing mislabeling errors in the training dataset.

SACUNet WS classification results (star markers in Fig. 4) reveal strong seasonal variations, with WS detections following expected trends at different elevations. The analysis of the correlation of SACUNet's WS detection with meteorological data will be provided in Section IV-B2.

There are some dates where discrepancies exist between the reference WS areas and the WS detected by SACUNet. These mismatches arise because, within a given patch of the reference WS map, in addition to reliably classified WS (represented by the bars in the figure) and reliably classified non-WS areas, there are also undefined regions. These uncertain areas result from cloud cover in the Sentinel-2 total snow product or data gaps.

3) Comparative Evaluation of DL Models: Fig. 8 provides a visual comparison of WS detection performance among different models trained using the 80×80 dataset. The proposed SACUNet outperforms the other models with small false alarm and missed alarm rates. In the examples shown in Fig. 8(a) and (b), the models perform well in accurately detecting large snow-covered areas. However, there is a noticeable issue with false alarms, where snow is overdetected in adjacent snow-free regions. This issue is considerably mitigated in the case of SACUNet. SACUNet also exhibits accurate detection of smaller snow patches [see example in Fig. 8(c)], as it is purposefully designed to capture and retain fine details from the original images. While these details are often identified and connected into larger snow patches, the resulting snow map tends to be overly regularized. This regularization leads to higher false alarm rates, particularly in the case of methods like SegNet, FCNet, or DeeplabV3+ when using backbones such

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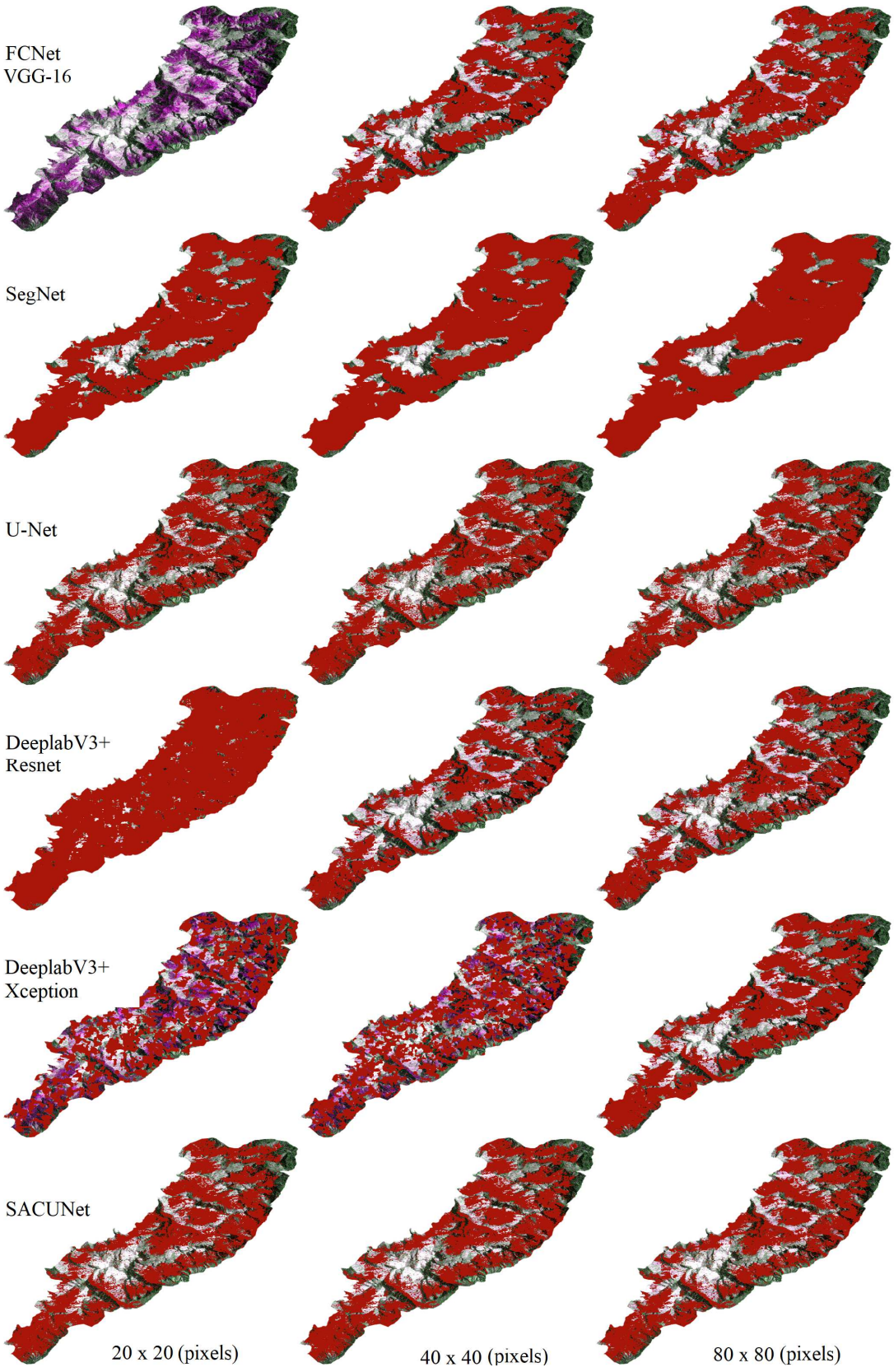


Fig. 9. WS (in red) on May 1, 2021, within the Mont-Blanc massif (Alps), overlaid on an RGB composite SAR image (R: January 19, 2020, G: May 1, 2021, and *B*: January 19, 2020). Each column corresponds to results obtained with the same input patch size (20×20, 40×40, and 80×80 pixels), while each row represents results from the same network architecture.

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TABLE II
PERFORMANCE EVALUATION OF DIFFERENT MODELS

Networks	Params (M)	OA (%)	Precision (%)	Recall (%)	IoU (%)	F1-score (%)
Input size 80×80 pixels						
FCNet VGG-16	134.2	91.55	86.21	87.79	76.99	87.00
SegNet	0.67	85.49	66.87	85.76	60.19	75.14
U-Net	31	93.99	91.52	90.28	83.32	90.90
Deeplabv3+ Resnet18	20.6	92.08	86.72	88.86	78.22	87.78
Deeplabv3+ Xception	27.6	91.06	82.96	89.04	75.27	85.89
SACUNet	29	97.45	94.44	97.72	92.40	96.05
Input size 40×40 pixels						
FCNet VGG-16	134.2	90.99	88.17	87.14	78.02	87.65
SegNet	0.522	86.04	71.27	87.97	64.94	78.74
U-Net	7.6	94.00	92.22	91.32	84.78	91.77
Deeplabv3+ Resnet18	20.6	91.83	87.14	90.03	79.47	88.56
Deeplabv3+ Xception	27.6	91.08	84.54	90.25	77.47	87.30
SACUNet	5.9	96.89	94.84	96.52	91.71	95.67
Input size 20×20 pixels						
SegNet	0.374	87.00	74.98	89.89	69.15	81.76
U-Net	1.8	94.08	93.04	91.84	85.93	92.43
Deeplabv3+ Resnet18	20.6	92.00	88.18	90.94	81.06	89.54
Deeplabv3+ Xception	27.6	91.59	88.88	89.41	80.42	89.14
SACUNet	1.7	96.89	95.79	96.18	92.29	95.99

as VGG-16, ResNet 18, or Xception. Small snow patches that persist for only a few days have been labeled as snow-free, as illustrated in the forest area in Fig. 8(d). This labeling approach relies on consistency among all three reference images and a corresponding Sentinel-2 total snow map for reliable WS labeling (see Section II-A). Consequently, snow pixels in such areas were not included in the sampling process. However, these pixels still exhibit a reduction in backscatter between the snow day and the reference day in log-ratio images, which is identified as WS by the U-Net and thus resulting in false alarms. The SACUNet also detected these details with a moderate level of regularization, while other models do not take these small patches of snow into account. It can be observed that missed detection rates are not particularly high for all methods, with the exception of SegNet [Fig. 8(e)].

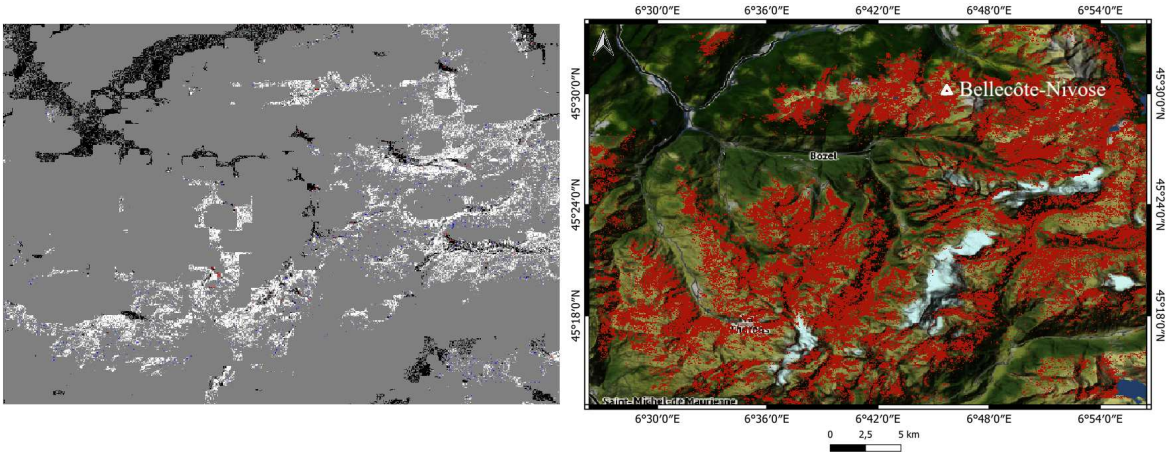
A quantitative evaluation of model performance, considering three distinct input image sizes of 80×80, 40×40, and 20×20 pixels, is detailed in Table II. Remarkably, SACUNet consistently demonstrates the highest performance across all three datasets, achieving an outstanding OA score of approximately 97%. These scores are accompanied by minimal error rates and notably high precision, recall, and $F1$-scores. U-Net exhibits commendable performance, while SegNet lags behind with lower precision and recall scores in comparison to the other models across all three datasets. Models employing very deep backbones for feature extraction, such as FCNet VGG-16, DeeplabV3+ ResNet, and Xception, do not perform optimally in the task of WS detection from medium-resolution SAR images like Sentinel-1, as evidenced by their comparatively lower scores. In general, the OA of each model exhibits little variation when utilizing different datasets for training. However, other metric scores tend to increase as the input data size decreases, primarily because more details are detected. Nevertheless, this can result in a somewhat noisier final output. As previously mentioned, the adaptability of SACUNet extends to the flexibility to adjust the number of encoder and decoder blocks. This adjustment aligns with the dimensions of the input data for sizes 80×80, 40×40, and 20×20 pixels. The number of parameters has decreased significantly, from 29M for input data size of 80×80 pixels to 5.9M for input data sizes of 40×40 pixels, and further down to 1.7 million for 20×20 pixel inputs. Consequently, not only does SACUNet exhibit remarkable accuracy, but it also optimizes computational resources by reducing the number of parameters to match the dataset's size. Among the tested methods, SegNet and U-Net are also adapted to the input data size, while the other methods are fixed.

In addition, Fig. 9 provides a comprehensive illustration of WS detection results, offering a visual correlation with the findings presented in Table II. The figure showcases the outcomes of WS detection across the entire Mont-Blanc massif on May 1, 2021, using the three distinct datasets. For a more intuitive comparison of the results, the detected WS regions were overlaid on the RGB composition SAR image, highlighting the areas with WS on the observed date, May 1, 2021, and the reference date, January 19, 2021. It was observed that an input size of 20×20 pixels is not compatible with the FCNet VGG-16 feature subsampling procedures. When employing input sizes of 40×40 and 80×80 for training, this model can detect WS in areas that correspond well with the background RGB image. However, the resulting WS maps appear significantly smoothed. Similar outcomes were obtained with the DeeplabV3+ ResNet model, where using a 20×20 pixel dataset resulted in an excessive detection of snow. For datasets of sizes 40×40 and 80×80 pixels, the results were quite similar to those of FCNet VGG-16. In the case of DeeplabV3+ Xception, the model exhibited issues with the misplacement of WS detection when trained with the 20×20 and 40×40 pixel datasets. Only when trained with the 80×80 pixel dataset did this model achieve WS detection

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TABLE III
WS DETECTION PERFORMANCE OF SACUNET ON THE VANOISE MASSIF TEST SITE

Input size (pixels)	OA (%)	Precision (%)	Recall (%)	IoU (%)	F1-score (%)
80 × 80	96.89	98.90	96.07	95.06	97.46
40 × 40	96.10	98.40	95.27	93.83	96.81
20 × 20	96.45	98.04	96.21	94.39	97.12



(a) (b)

Fig. 10. WS detection of SACUNet over the test site of Vanoise massif. (a) SACUNet's performance: white—true positive, black—true negative, red—false positive, and blue—false negative. (b) Detected WS (in red, overlaid on Stamen Terrain map) over the Vanoise massif using SACUNet trained on the 80 × 80 pixel input size Mont-Blanc dataset. White triangle: position of Bellecôte-Nivose meteorological station (2992 m).

results, but with a high degree of regularization. SegNet experienced issues with over-detection with all three training datasets, marking almost every area as snow-covered. U-Net and SACUNet yielded relatively similar results. However, unlike U-Net, SACUNet was less affected by small details and transient changes of snow cover, by moderately regularizing the obtained results while preserving the original image's features effectively.

4) Cross-Site Generalization to the Vanoise Massif: To demonstrate the effectiveness of SACUNet's WS detection capabilities, we extended its evaluation beyond the Mont-Blanc massif to include the Vanoise massif as a test site. The results of this evaluation, as depicted in Table III, reflect very good performance metrics similar to those observed in the Mont-Blanc region. Fig. 10 offers a visual insight into the WS detection process. Specifically, Fig. 10 shows the detection errors of the SACUNet model trained with the 80 × 80 pixel Mont-Blanc dataset, and Fig. 10(b) exhibits the resulting WS map for the Vanoise massif on June 6, 2021. This assessment revealed small errors, primarily occurring in the transitional zones between WS and snow-free areas. Consequently, the delineation of WS proved to be notably accurate. It is worth noting that, for practical applications in mountainous regions, the WS map in Fig. 10(b) with nonmountainous areas was masked out using data from Copernicus DEM GLO30 and CORINE Land Cover 2018, ensuring a more focused and informative representation of WS distribution in mountainous terrains. Additionally, the effects of radar-specific imaging geometric distortions, including foreshortening, foldover, and shadow, were also masked. We applied these masks to all WS product maps (e.g., extent maps and frequency maps) presented throughout the study's illustrations going forward.

SACUNet WS detections at Bellecôte station (2992 m, Vanoise massif) in Fig. 5 show clear seasonal variability, strongly aligned with observed temperature, snow depth, and humidity trends. Throughout the observation period, SACUNet effectively captured the seasonal evolution of WS, with frequent detections during summer (July–August 2020 and 2021), a gradual decline through autumn (mid-August–November), minimal presence in winter (December 2020–February 2021), and a progressive increase in spring (March–June), peaking in May–June. These results confirm the model's capacity to track WS dynamics in response to environmental forcing. Temperature exerts the dominant control: WS appears when maximum temperatures exceed 0 °C, especially in spring and summer, and nearly disappears under persistent subzero winter conditions. Snow depth also influences detection, decreasing WS presence during early winter accumulation and increasing it again in spring as snow depth declines with melting. Humidity plays a secondary but consistent role; high humidity (>80%) generally coincides with WS detections in warm periods, suggesting that moisture retention enhances radar wetness signals. Several short-term events highlight SACUNet's sensitivity to environmental transitions. In early autumn (September 2020 and 2021), fluctuations in WS detections corresponded to brief warm episodes, likely linked to melt-refreeze cycles. In late winter (February 2021), a slight increase in detections, despite low air temperatures, may result from localized warming, transient snowpack changes, or surface roughness effects. The largest WS coverage in May 2021 coincides with higher temperatures, rising humidity, and rapid snow depth decrease,

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TABLE IV

EXAMPLES OF CORRECT WS DETECTION CASES AT THE BELLECÔTE STATION (VANOISE MASSIF). EACH ROW LISTS SACUNET WS DETECTIONS ALONGSIDE METEOROLOGICAL CONDITIONS (TEMPERATURE, SNOW DEPTH, AND HUMIDITY). THERE ARE EXAMPLES THAT DEMONSTRATE AGREEMENT BETWEEN MODEL PREDICTIONS AND EXPECTED WS PRESENCE (WHEN TEMPERATURE > 0 °C, SNOW DEPTH > 10 CM, AND HUMIDITY ≥ 90%) OR ABSENCE (WHEN TEMPERATURE < 0 °C FOR > 48 H AND HUMIDITY ≤ 70%

Date	WS (%)	Tmax (0°C)	Snow Depth (cm)	Humidity (%)	Expected Condition	WS Detected	Validation Result
2020-07-11	34.5	5.1	177.0	90.0	Expect Wet Snow	Yes	Correct
2020-08-04	29.5	6.6	25.0	96.0	Expect Wet Snow	Yes	Correct
2020-11-02	36.3	6.0	74.0	90.0	Expect Wet Snow	Yes	Correct
2020-12-26	4.8	-5.6	140.0	62.0	Expect No Wet Snow	No	Correct
2021-01-19	0.0	-5.0	193.0	47.0	Expect No Wet Snow	No	Correct
2021-02-06	0.4	-0.6	288.0	62.0	Expect No Wet Snow	No	Correct
2021-02-18	1.7	-2.2	273.0	43.0	Expect No Wet Snow	No	Correct
2021-03-26	3.8	-1.4	290.0	35.0	Expect No Wet Snow	No	Correct
2021-04-13	7.6	-7.5	285.0	55.0	Expect No Wet Snow	No	Correct
2021-08-05	14.3	4.5	18.0	91.0	Expect Wet Snow	Yes	Correct

confirming SACUNet's accuracy in capturing the peak melt season.

Table IV shows a subset of example cases where SACUNet predictions align with these simple rules.

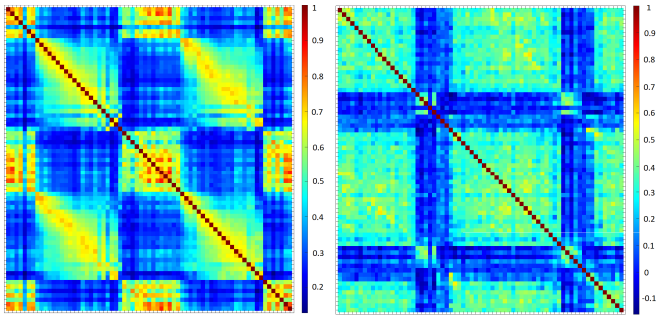
1) WS is expected when temperature > 0 °C, snow depth >10 cm, and relative humidity ≥ 90%.
2) No WS is expected when temperature < 0 °C for more than 48 h and relative humidity ≤ 70%.

The findings indicate that SACUNet is sensitive to seasonal variations and adapts well to local weather-driven snowpack conditions.

Overall, SACUNet exhibits high reliability in detecting WS at Bellecôte station, with strong agreement between model outputs and meteorological observations. Importantly, although SACUNet was trained using a database constructed in the Mont-Blanc massif, it successfully detected WS in the Vanoise massif. This demonstrates the ability of the proposed method to generalize across different mountainous regions, beyond the area where the training data were derived. These results confirm SACUNet's robustness in monitoring WS dynamics and reinforce its suitability for operational WS mapping in alpine environments.

B. WS Monitoring in the Mont-Blanc Massif

1) Selection of Reference Dates: Selecting appropriate reference images is essential for accurate WS detection from SAR data. The reference scene must correspond to a snow-free surface so that backscatter variations reflect only snowmelt effects. Following the correlation-based approach proposed by Karbou et al. [41] and Karas et al. [70], we analyzed temporal correlation matrices across representative zones of the Mont-Blanc massif (Fig. 11). The first site, located at about 3000 m (see Fig. 3), exhibits clear seasonal patterns, with high correlations (0.7–0.8) during stable dry- or wet-snow periods and weaker values (0.3–0.5) during transitional phases, illustrating the gradual evolution of surface conditions between summer and winter. In contrast, the glacierized area near 2000 m shows lower and more variable correlations (−0.1 to 0.6) due to glacier motion and surface instability, which affect radar backscatter independently of snow conditions. Notably, the Argenti Ãre Glacier can move up to 20 cm/day [71], making its surface intrinsically unstable regardless of snow cover.



(a) (b)

Fig. 11. Matrix depicting correlation coefficients among images within the time series. (a) Correlation matrix of area 1 (marked in Fig. 3) situated at an elevation of approximately 3000 m near the Mont-Blanc mountain. (b) Correlation matrix of area 2, Miage glacier (marked in Fig. 3), located at an elevation of roughly 2000 m.

Across the study area, correlation stability is typically observed from January to March and from July to August, corresponding, respectively, to dry snow and melt seasons. Based on this analysis, three winter dates—January 13, 19, and 25, 2021—were selected as reference scenes, representing periods when WS was confidently absent and surface backscatter remained stable.

2) WS Extent, Progression, and Frequency: For the monitoring of WS in the Mont-Blanc massif, the trained SACUNet model was applied to every image in the image time series to determine the presence of WS from each Sentinel-1 observation. Fig. 12 shows examples of obtained WS maps over the Mont-Blanc massif for every month of the observation period and the area of WS detected (in ha) for every observation from July 5, 2020, to August 29, 2021, which allows the monitoring of WS extent and progression. The Mont-Blanc massif is characterized by high elevations, and temperature decreases with altitude. Analyzing the snow extent maps for each date in the observation time series and correlating the distribution of WS with altitude reveals several notable patterns. During the summer months of July and August 2020, WS was consistently present at higher altitudes, ranging from 2700 to well over 4000 m. This persistence could be attributed to the spring thaw and the accumulation of snow at these altitudes. In fact, WS was still observed at altitudes exceeding 4200 m during this season. However, as September 2020 began, a gradual

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Fig. 12. Examples of SACUNet's WS extent (in red, overlaid on Stamen Terrain map) mapping in the Mont-Blanc massif of the Alps.

decrease in the WS extent unfolded at altitudes around 4000 m. WS was detected from an altitude of 2800 m and above. With the progression of autumn and decreasing temperatures, WS diminished further. From mid-September onward, WS

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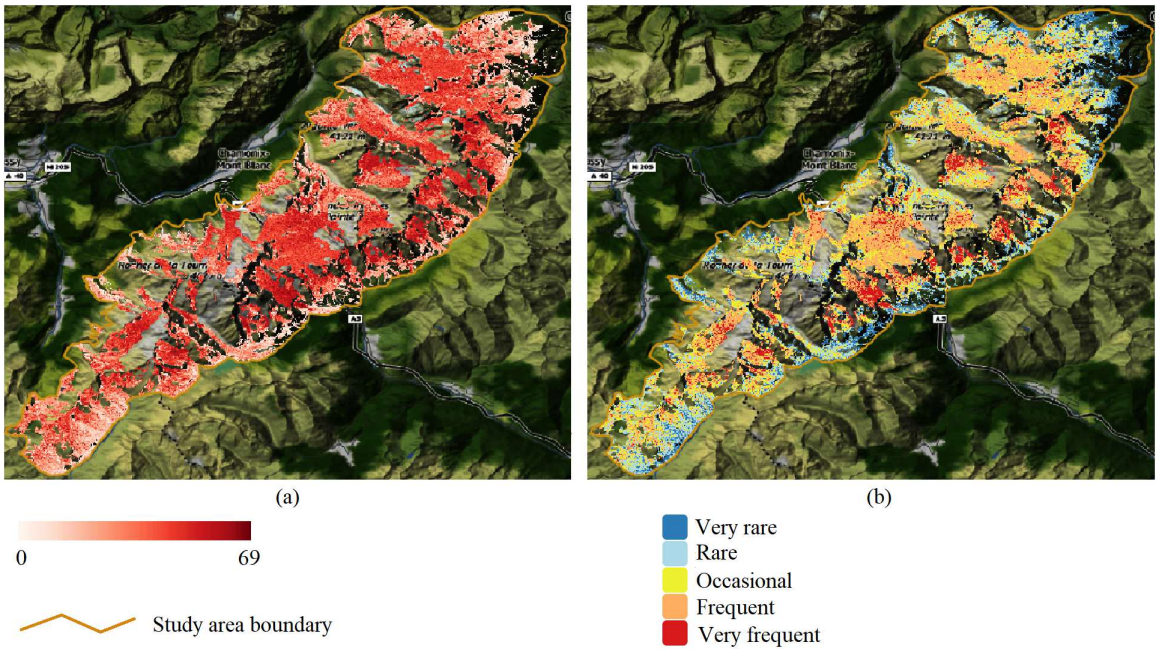


Fig. 13. WS frequencies (overlaid on Stamen Terrain map) in the Mont-Blanc massif during the observation period from July 5, 2020, to August 29, 2021. (a) Cumulative occurrence of WS. (b) WS frequency.

was primarily observed between altitudes of 2900 and 3900 m; at lower altitudes, there was very little WS observed; at higher altitudes, the snow has become dry. In October and November 2020, snow extent gradually receded. The arrival of winter (December 2020–February 2021) brought a significant drop in temperature, leading to a marked reduction in WS occurrences. The lowest WS extent was recorded in January 2021, consistent with the coldest temperatures of the year.

These results are further validated by in situ meteorological observations from the three stations, Aiguille du Midi (3842 m), Plan de l'Aiguille (2317 m), and Lognan (1970 m) (Fig. 4). SACUNet's detections at different elevations align with recorded temperature fluctuations, total snow depth, and humidity levels. Notably, in late winter (February 6, 2021), a large WS area was detected at Lognan, despite overall cold conditions, likely due to localized warming and transient liquid water retention within the snowpack. Similarly, on April 1, 2021, a sharp increase in WS detections at Plan de l'Aiguille and Lognan coincided with a spring warming event, marking the transition into seasonal melt. WS was also detected at Aiguille du Midi in late spring (May–June 2021), confirming the presence of high-altitude melting events, where SACUNet successfully distinguished persistent dry snow from transient wet conditions. The overall consistency between SACUNet WS classifications and meteorological trends confirms that the model accurately tracks melting cycles, with minimal false detections during colder months.

In general, SACUNet's WS classification across the Mont-Blanc massif effectively captures seasonal transitions and elevation-dependent melting patterns. The model demonstrates high classification accuracy, as validated by in situ meteorological observations at multiple stations. The detection of key seasonal transitions, including the sharp rise in WS in spring, peak melting in early June, and near-total absence of WS in midwinter, confirms SACUNet's reliability. Although the lack of meteorological data from July to December 2020 limits full validation for that period, the observed alignment of WS detection with known seasonal trends further reinforces SACUNet's effectiveness in WS monitoring across complex mountainous environments.

Fig. 13 presents the WS duration and frequency maps derived from the full time series of WS extent. The cumulative map in Fig. 13(a) shows the total number of WS occurrences, with a maximum index of 69 (corresponding to the number of Sentinel-1 observations). Fig. 13(b) classifies WS frequency into six levels, from areas with no detections to those showing very frequent WS cover. These maps provide an overall view of WS distribution and recurrence across the Mont-Blanc massif. Regions with very rare and rare WS occurrences are mainly found between 1600 and 2400 and above 4000 m, including glacier zones and outer contours of the study area. These account for 10.14% and 21.01% of the total WS-covered area, respectively. Occasional WS frequencies represent about 27.02%, mostly located in the southwest sector, glacier regions, and the central ridge of Mont-Blanc, spanning 2000–4000-m elevations. Areas with frequent WS occurrences cover 32.72%, dominating the central massif within similar altitude ranges. Finally, very frequent WS areas make up 9.11% of the total, scattered mainly along the eastern edge and locally in the central massif.

3) WS Detection Result Comparison: To enhance the reliability of the proposed method, we also conducted a comparative analysis between the WS detection results obtained from the SACUNet and Copernicus WS maps. This comparison was performed using reference dates selected from two different seasons of the year. It is worth noting that the Copernicus WS maps were generated from C-band Sentinel-1 images based on detecting the reduction of backscatter coefficient (indicating WS) through the ratio between the SAR

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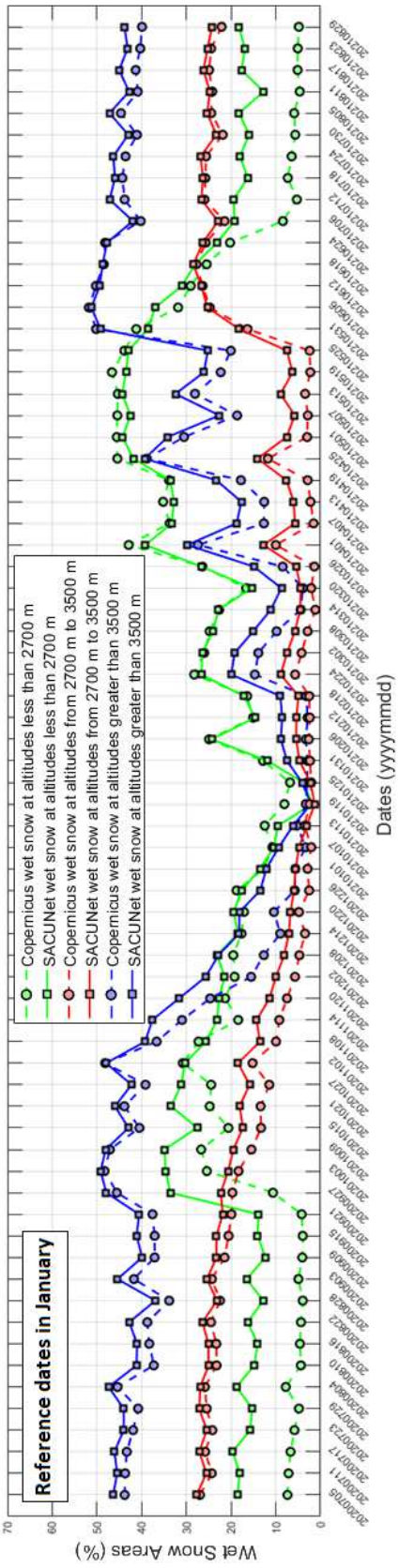


Fig. 14. Percentage of WS pixels detected by SACUNet and those on Copernicus maps at three elevation ranges over time in the Mont-Blanc massif during the observation period from July 5, 2020, to August 29, 2021, with reference dates selected in January.

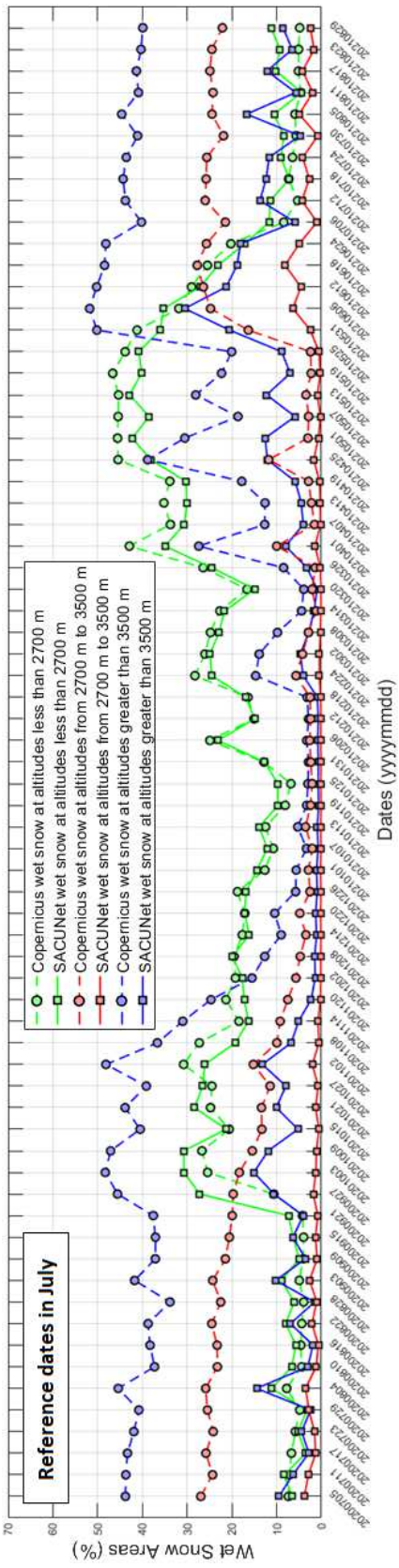


Fig. 15. Percentage of WS pixels detected by SACUNet and those on Copernicus maps at three elevation ranges over time in the Mont-Blanc massif during the observation period from July 5, 2020, to August 29, 2021, with reference dates selected in July.

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image of the observation date and the SAR image of the reference date when no snow was present.

First, January 13, 19, and 25, 2021, were selected as three winter reference dates. Fig. 14 illustrates the percentage of WS pixels detected by SACUNet and those indicated on the Copernicus maps, stratified by three elevation ranges over time, i.e., below 2700 m, between 2700 and 3500 m, and above 3500 m, which represent 51.52%, 40.16%, and 8.32% of the total area of the Mont-Blanc massif, respectively. The WS area graphs from SACUNet (solid lines) and Copernicus (dashed lines) are notably consistent, particularly for altitudes above 2700 m. From 2700 to 3500 m, significant WS was observed in the summer months of July, August, and September 2020, ranging from 40% to 50%. Subsequently, WS gradually diminished from October 2020, reaching minimal levels in the middle of winter, specifically January and February 2021. The amount of WS gradually increased, peaking at 51.26% in early June 2021, and maintaining substantial amounts throughout the summer. Patterns of the graphs at elevations above 3500 m closely resembled those observed in the 2700–3500-m range, though with lower amplitudes. During the winter months, from December 2020 to the end of March 2021, almost no WS was observed.

At altitudes below 2700 m, however, seasonal trends and agreement between SACUNet and Copernicus diverged. Specifically, minimal WS was observed in the summer, from July to mid-September in 2020 and 2021, registering less than 20% for the SACUNet method and less than 10% for the Copernicus map due to melted snow. This difference between SACUNet and Copernicus results arises because the WS detected at this elevation range in this period was predominantly patchy and was masked on Copernicus maps. By September 27, 2020, the WS amount surged by 20% compared to the previous day, from 13.89% to 33.61%, possibly due to heavy rain. After that, WS gradually decreased, reaching a minimum of 3.28% in mid-January 2021. Between January 20, 2021, and May 25, 2021, when WS coverage was increased over time, the area of WS detected at elevations below 2700 m was larger than that of the two higher elevation ranges. From November 2020 to the end of June 2021, the WS graphs from SACUNet and Copernicus exhibited high similarity.

Subsequently, reference dates were chosen during the summer, specifically July 5, 17, and 29, 2020. As shown in Fig. 15, using these summer reference dates led the SACUNet method to detect minimal WS across all elevations during the summer months. Above 2700 m, WS detections remained sparse for the remainder of the observation period, with significant discrepancies between the SACUNet and Copernicus methods. In contrast, at elevations below 2700 m, the SACUNet and Copernicus results showed strong agreement, indicating consistent winter WS detection from October 2020 to June 2021 below 2700 m. Additionally, the spatial distribution of WS during the summer months (July–September) differed notably from the patterns observed between October 2020 and June 2021. These observations support the choice of January reference dates as more suitable for this study area.

C. Discussion

As mentioned earlier, the lack of independent ground-truth data poses challenges for validating our approach comprehensively. This constraint is particularly relevant in complex alpine environments where access to the study area is difficult. As a result, obtaining consistent, on-the-ground measurements for validation is challenging, leading us to rely primarily on high-confidence satellite-derived samples, confirmed across multiple reference products, for labeling. While this method minimizes some validation gaps, the absence of independent datasets may affect generalizability, particularly across regions with differing snow or surface conditions. Thus, while our methodology provides a robust framework for snow detection, incorporating independent ground-truth data in future work would further strengthen the accuracy and reliability of our results, particularly in complex or variable snow conditions.

Another challenge arises in glacierized regions, where WS detection is complicated by the presence of bare ice beneath dry snow and by glacier surface instability, both of which reduce the reliability of reference scenes. Unlike stable terrain, glaciers undergo continuous ice flow, crevasse formation, and seasonal snowpack variations, causing backscatter variability that affects WS classification. Additionally, exposed ice surfaces beneath thin snow layers can mimic WS signatures, increasing misclassification risks. To mitigate this, we prioritized midwinter reference scenes (e.g., January), when glaciers are most consistently covered by dry snow, reducing spectral distortions. However, snowpack thickness variations and superimposed ice layers still pose classification challenges. As shown in Fig. 16, WS evolution over glacier areas varies strongly with altitude. At lower elevations (<2300 m, e.g., Mer de Glace and Argentiére), WS follows a temperature-driven seasonal cycle, peaking at 40%–50% in autumn, disappearing by January, and reappearing rapidly in spring (70%–80%) during melting. Detection there is generally reliable, while higher elevations face greater uncertainty due to glacier instability and bare-ice exposure affecting backscatter. At mid-altitude glaciers (~3000 m, e.g., Bossons Glacier), WS remains more persistent, 70%–80% in summer with a gradual autumn decline, and about 10% persists through winter, likely from melt-refreeze cycles or trapped water. Bare ice beneath seasonal snow increases backscatter variability, complicating the distinction from ice-related signals. Multitemporal analysis mitigates these effects by emphasizing consistent seasonal patterns. At higher altitude glaciers (~4000 m, e.g., Taconnaz Glacier), WS is stable (>80% in summer and ~40% in winter), likely due to snow insulation or solar heating on steep slopes. However, glacier motion and roughness induce strong backscatter variability, challenging classification. Overall, glacierized areas follow the Mont-Blanc trend but retain more WS (~40% at 4000 m) due to subsurface ice, radiation effects, and snow insulation. WS covers 25%–30% in summer, rises to 34%–40% in autumn, and drops to 1.9% in midwinter. Distinct surges appear after major weather events (e.g., September 27, 2020, and May 31, 2021). While multitemporal training reduces errors, glacier variability still drives uncertainty, suggesting future work should combine in situ and unmanned aerial vehicle (UAV) validation, multifrequency SAR (L-band and X-band), glacier masks, and uncertainty assessments.

An additional source of uncertainty arises from the binary classification adopted in this study, where all dry

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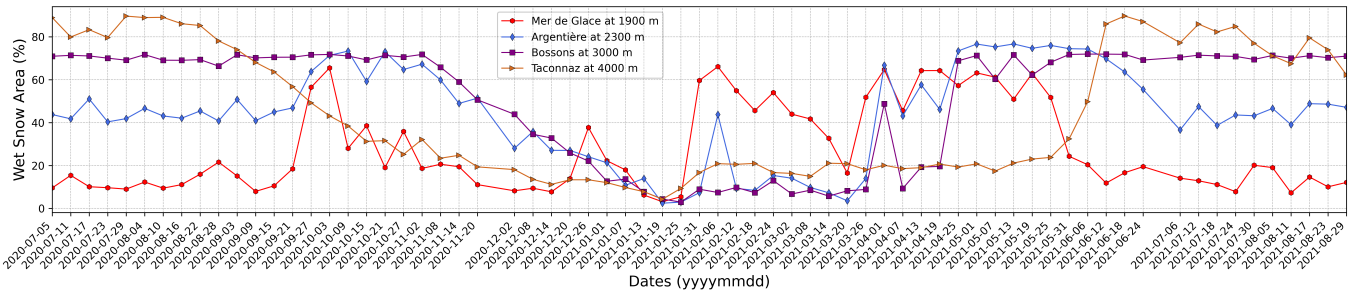


Fig. 16. Examples of WS evolution at glacier areas spanning the observation period from July 5, 2020, to August 29, 2021, at different altitudes.

snow is grouped with snow-free terrain as "non-WS." This simplification may introduce misclassifications because dry snow can sometimes exhibit radar backscatter similar to WS, depending on local environmental conditions. The impact of this choice cannot be explicitly quantified with the current dataset, as no large-scale reference exists to distinguish between dry snow and WS across the Alps. Nonetheless, this limitation points toward future research directions. One natural extension is to move toward a three-class framework (WS, dry snow, and snow-free). This could be achieved either by training a more complex model on explicitly labeled datasets, once sufficient reference data become available, or by postprocessing binary WS outputs with auxiliary information such as meteorological thresholds or independent optical snow products (e.g., VIIRS). Such approaches could reduce ambiguities in snow state separation without requiring large-scale ground-based campaigns.

Building on the encouraging results, several directions could further enhance the framework. Integrating environmental drivers such as temperature, humidity, and elevation into the learning process would improve robustness in complex alpine regions, where terrain and microclimate strongly modulate melt dynamics. Complementary optical snow products (e.g., Sentinel-2 and Pléiades) could serve as independent References for cross-validation, strengthening confidence in model outputs, particularly where in situ data are lacking. Expanding evaluation beyond Mont-Blanc and Vanoise to other mountain ranges with available validation datasets would also provide insight into the generalizability of SACUNet across diverse snow regimes. Looking further ahead, extending classification toward finer snow states, such as distinguishing melt phases described in [25], will benefit from future SAR constellations with higher revisit frequencies, which are crucial for capturing rapidly evolving snowpack conditions.

V. Conclusion

In this article, we have proposed a deep semantic learning framework designed for the segmentation of Sentinel-1 images, aimed at the detection and monitoring of WS in mountainous regions. To tackle the challenge of mapping WS in high mountainous terrains like the Mont-Blanc massif, we have constructed comprehensive databases for model training and validation. These databases rely on validated snow products derived from both Sentinel-1 SAR images and Sentinel-2 optical images. We also introduced two quantitative metrics, CAR and WSIoU, to explicitly evaluate the fusion-based label generation process. Their joint use highlighted both the strengths and the limitations of the fusion process, offering a robust statistical framework to validate training label reliability in the absence of direct ground truth. The proposed network, SACUNet, exhibits remarkable accuracy in the detection of WS within mountainous areas with performance metrics including an OA of 97%, precision of 94%, recall of 97%, IoU at 92%, and $F1$-score reaching 96%. Furthermore, we have conducted comparative evaluations of our method against several state-of-the-art models, such as the FCNet VGG16, SegNet, U-Net, DeeplabV3+ ResNet18, and DeeplabV3+ Xception, and SACUNet consistently outperforms these models. To extend the applicability of the proposed model, we have put SACUNet to the test by applying the model, trained on databases created for the Mont-Blanc massif, to the task of detecting WS in the Vanoise massif. We achieved similarly high accuracy in generating WS maps for this new region.

This article also analyzed WS monitoring in the Mont-Blanc massif for a year. The analysis is based on the results of the SACUNet model, which used a series of 69 SAR images taken between July 5, 2020, and August 29, 2021. The analysis provides insight into the extent, patterns, frequency, and temporal evolution of WS in the high-altitude Mont-Blanc massif in the Alps. Moreover, meteorological observations (temperature, snow depth, and humidity) from stations at different elevations were used to confirm the seasonal consistency of SACUNet detections, reinforcing confidence in the physical consistency of the results.

The proposed framework demonstrates significant potential for accurate and scalable WS detection, offering a robust foundation for future improvements and broader applications in alpine and other snow-dominated regions although the lack of reliable ground-truth testing data may introduce certain limitations.

Appendix
SACUNet: Input–Output Relationships for the Feature Maps Involved in the Model

The entry block takes an input data $\mathcal{S}$ defined as in (2) and it outputs the feature image

$$\left(J^0_{\bullet,\bullet,q}\right)_{1\leqslant q\leqslant N_0} = \mathsf{f}\left[\mathsf{g}\left(\left(\mathcal{S}_{\bullet,\bullet,k}\right)_{1\leqslant k\leqslant k_0} \circledast \mathsf{H}^{3,3,K}_{N_0,1,1}\right)\right] \tag{11}$$

where g is the batch normalization operator, and f denotes the ReLU function.

The output feature from J^{u+1} of a decoder block with respect to the input J^u is the following addition associated

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with two parallel branches described in the block diagram on the left of Fig. 2(b):

$$\left(J^{u+1}_{\bullet,\bullet,q}\right)_{1\leqslant q\leqslant N_{u+1}} = \ell_2^{3,3}\Big[\mathsf{g}\Big(\Big[\Lambda\Big(\Big[\mathsf{f}_u\big(\mathrm{J}^u\big)\circledast \mathbf{H}^{3,3,1,N_u}_{1,1,1}\Big] \circledast \mathsf{H}^{1,1,1}_{N_{u+1},1,1}\Big)\circledast \mathbf{H}^{3,3,1,N_{u+1}}_{1,1,1}\Big] \circledast \mathsf{H}^{1,1,1}_{N_{u+1},1,1}\Big)\Big] + \mathsf{g}(J^u \circledast \mathsf{H}^{1,1,1}_{N_{u+1},2,1}), \tag{12}$$

where

$u \in \{0,1,\ldots,U\}$	different encoder blocks;
N_u	desired number of filters for the pointwise convolutions, with $N_{u+1} = 2N_u$;
$\Lambda = \mathsf{f}\circ\mathsf{g}$,	where we considered f_0 = Identity and $\mathsf{f}_1 = \mathsf{f}_2 = \mathsf{f}_3 = \mathsf{f}$;
$\ell_s^{M_1,M_2}$	maxpooling operator with size $M_1 \times M_2$ and stride s.

The feature image $(J^{U+2}_{\bullet,\bullet,q})_{1\leqslant q\leqslant N_{U+2}}$ of the atrous block follows from:

$$\begin{aligned}\left(J^{U+2}_{\bullet,\bullet,q}\right)_{1\leqslant q\leqslant N_{U+2}} = \mathsf{f}\Big[&\mathsf{g}\big(J^{U+1}\circledast \mathbf{H}^{1,1,1,N_{U+1}}_{1,1,1}\circledast \mathsf{H}^{1,1,N_{U+1}}_{N_{U+1}/4,1,1}\big)\\ \|&\mathsf{g}\big(J^{U+1}\circledast \mathbf{H}^{2,2,1,N_{U+1}}_{1,1,2}\circledast \mathsf{H}^{1,1,N_{U+1}}_{N_{U+1}/4,1,1}\big)\\ \|&\mathsf{g}\big(J^{U+1}\circledast \mathbf{H}^{2,2,1,N_{U+1}}_{1,1,3}\circledast \mathsf{H}^{1,1,N_{U+1}}_{N_{U+1}/4,1,1}\big)\\ \|&\,\mathsf{g}\big(J^{U+1}\circledast \mathbf{H}^{2,2,1,N_{U+1}}_{1,1,4}\circledast \mathsf{H}^{1,1,N_{U+1}}_{N_{U+1}/4,1,1}\big)\Big]\end{aligned} \tag{13}$$

with $\bullet \,\|\, \bullet$ representing the concatenation operation.

For the upper index $V = U + 3$ in J, we have a bridge in SACUNet that results in

$$\left(J^{V}_{\bullet,\bullet,q}\right)_{1\leqslant q\leqslant N_{U+3}} = \mathsf{f}\left(J^{U+2}\circledast \mathsf{H}^{3,3,N_{U+1}}_{N_{U+3},1,1}\right). \tag{14}$$

After this bridge, the output of each decoder block for $v \in \{1,2,\ldots,U\}$ is computed from

$$\left(J^{V+v}_{\bullet,\bullet,q}\right)_{1\leqslant q\leqslant N_{V+v}} = \mathsf{f}\Big[\mathsf{f}\Big(\Big[\mathsf{f}\Big(J^{V+v-1}\circledast \mathsf{T}^{3,3,N_{V+v-1}}_{N_{V+v},2,1}\Big) \,\|\, J^{\text{V-2-v}}\Big]\circledast \mathsf{H}^{3,3,N_{V+v}}_{N_{V+v},1,1}\Big)\circledast \mathsf{H}^{3,3,N_{V+v}}_{N_{V+v},1,1}\Big] \tag{15}$$

where T denotes a transposed convolution tensor when using the same index conventions as for the direct convolution, and $N_{V+v} = N_{V+v-1}/2$.

The last feature map is then derived from

$$\left(J^{V+U+2}_{\bullet,\bullet,q}\right)_{1\leqslant q\leqslant N_{V+U+2}} = J^{V+U+1}\circledast \mathsf{H}^{3,3,N_{V+U+1}}_{N_{V+U+2},1,1}. \tag{16}$$

Specifically, $N_{V+U+2} = 2$ to specify a binary output.

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